CS7646 / MACHINE LEARNING FOR TRADING

Make decisions. Measure the edge. Price the trade.

Exam 2 | Machine learning + quantitative finance

A printable companion to the Study Lab: sequential decision-making, reinforcement learning, trading signals, evaluation, execution, portfolio construction, and options.

420 questions	210 machine learning + 210 quantitative finance
140 groups	70 ML groups + 70 quantitative finance groups
20 domains	10 ML chapters + 10 quantitative finance chapters
Source scope	ML Part 2: pool pp. 421-581 / QF Part 2: pp. 582-753

The study objective

Explain the mechanism, perform the calculation, and interpret the result under the stated assumptions. Then check whether the item uses ordinary or reversed True/False labels.

Built around the supplied ML4T Exam Question Pool, Rev081020261230. All source references use 1-based PDF pages. This is a teaching synthesis; coverage counts do not predict exam frequency. No official exam deadline was supplied.

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How to use this guide

Read a domain, close the guide, and explain its rules from memory. Reproduce the worked calculations before looking at the result. Use the final mixed self-test to locate weak concepts, then return to the linked source question IDs.

Read the stem first. 90 of the 420 source items explicitly reverse the convention: mark True when the claim is incorrect and False when it is correct. Evaluate the underlying claim before translating it into the requested label.

Track the assumptions. Use the stated reward timing, terminal-state treatment, exploration rule, return horizon, fee convention, cost base, and payoff-versus-profit definition. Nested negation requires evaluating both the claim and its wrapper.

Source cautions. Published labels are retained in the Study Lab. A flagged contradiction is not silently corrected: learn the defensible concept and record the published selection separately. The caveat section identifies known conflicts and qualifications.

Domain directory

MACHINE LEARNING	QUANTITATIVE FINANCE
ML2-D1 Reinforcement Learning Problem Framing	**QF2-D1** Financial Time Series and Market Data Behavior
ML2-D2 Markov Decision Processes, Policies, and Values	**QF2-D2** Technical Indicators and Signal Construction
ML2-D3 Q-Learning Mechanics	**QF2-D3** Alpha Research and Formulaic Signals
ML2-D4 Exploration, Exploitation, and Learning Dynamics	**QF2-D4** Fundamental Law of Active Portfolio Management
ML2-D5 Dyna and Model-Based Reinforcement Learning	**QF2-D5** Advanced Portfolio Construction
ML2-D6 State Representation, Discretization, and Reward Design	**QF2-D6** Strategy Evaluation and Performance Attribution
ML2-D7 Reinforcement Learning for Trading	**QF2-D7** Trading Frictions, Capacity, and Implementability
ML2-D8 Comparative Strategy Learning and Evaluation	**QF2-D8** Execution Algorithms and Order Book Analytics
ML2-D9 Deep Learning and Neural-Network Foundations	**QF2-D9** Options and Derivative Payoffs
ML2-D10 Sequential, Generative, and Responsible AI	**QF2-D10** AI-Enabled Investment Processes and Decision Support

Formula sheets (start p. 3) / 20-question self-test / Separate answers / Source cautions

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Formula sheet | Machine learning

CONCEPT	FORMULA AND CONDITIONS
Discounted return	$G_t = r_{t+1} + \gamma r_{t+2} + \gamma^2 r_{t+3} + \ldots$ γ controls discounting of later rewards; it is not the learning rate.
One-step Q-learning	$\text{target} = r + \gamma \max_a Q(s', a)$ $Q_{new}(s,a) = Q_{old}(s,a) + \alpha[\text{target} - Q_{old}(s,a)]$ Use the next-state maximum, not the current-state maximum. Only the visited state-action entry is directly updated.
Terminal update	$\text{target} = r$; $Q_{new} = (1-\alpha)Q_{old} + \alpha r$ A true terminal transition has no continuation value. Artificial truncation and true termination need not be equivalent.
Epsilon-greedy	$P(a_{greedy}) = 1-\varepsilon+\varepsilon/k$ $P(a_{other}) = \varepsilon/k$ Assumes one greedy action and uniform exploration over all k legal actions. If random exploration excludes the greedy action, probabilities differ.
Exploration decay	$\varepsilon_n = \varepsilon_0 \times d^n$ Counts n decay applications. Declining exploration changes action selection; it does not itself reduce the Q-learning step size.
Dyna planning	real experience + model updates + simulated Q updates The same temporal-difference rule can update real and simulated transitions. More planning improves reuse but can amplify model error.
Discrete state space	$N_{states} = \prod b_j$; $N_{Q_entries} = N_{states} \times N_{actions}$ Multiply state-variable cardinalities. Include holdings or time when they change reward or feasible future actions.
Target position to trade	$\text{trade}_t = \text{target_holdings}_t - \text{current_holdings}_t$ Moving from +L to -L trades -2L. A target of zero liquidates an existing holding; it is not automatically “do nothing.”
Returns and sample evaluation	$r_t = V_t/V_{t-1} - 1$ $R = \prod(1+r_t) - 1$ Use aligned dates, capital, execution, leverage, and costs. Repeated historical episodes are not independent new market samples.
Neural layer	$h = \text{activation}(Wx+b)$ Nonlinear activations allow nonlinear compositions. Extra parameters or channels do not establish causality or generalization.

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Formula sheet | Quantitative finance

CONCEPT	FORMULA AND CONDITIONS
Trailing indicators	SMA(n) = mean(last n prices) Momentum(L) = P_t/P_(t-L) - 1 EMA_t = αP_t + (1-α)EMA_(t-1) A common EMA convention uses α=2/(n+1). Specify initialization, window, and information timestamp.
Bands and normalized position	Upper / Lower = μ ± kσ z = (P-μ)/σ %B = (P-Lower)/(Upper-Lower) A band breach is an observation. %B can exceed [0,1]; neither a band touch nor “overbought” guarantees reversal.
Information coefficient	IC = Corr(forecast, subsequent outcome) Align observations, horizon, and information time. Rank IC measures rank agreement; magnitude and economic value are different questions.
Fundamental Law	IR ≈ IC × sqrt(BR) Constrained extension: IR ≈ TC × IC × sqrt(BR) BR is effective independent breadth, not raw correlated trade count. TC represents implementation fidelity under the stated extension.
Sharpe and information ratio	Sharpe = mean(r_p-r_f)/sd(r_p-r_f) IR = mean(r_p-r_b)/sd(r_p-r_b) Use matched periodicity. Under suitable assumptions, annualize a periodic ratio by sqrt(periods/year). Denominators differ.
Trading costs	cost = traded notional × bps/10,000 net profit = gross profit - all modeled costs Check one-way versus round-trip quotes and turnover convention. Spread, commission, impact, delay, and opportunity cost can overlap if carelessly combined.
Option expiration payoff	Call = max(S_T-K, 0) Put = max(K-S_T, 0) Long-option profit subtracts the premium paid; short-option profit is the premium received minus its payoff obligation.
Option breakeven	Long call: K + call premium Long put: K - put premium Simple expiration profit, per share, before financing, dividends, fees, and contract multiplier effects.
Put-call parity	C - P = S_0 - PV(K) Same strike and expiration, European exercise, and stated no-dividend/no-friction assumptions. Adjust the identity when assumptions change.
Active portfolio construction	active weights a = w-b variance = $w^T\Sigma w$; $TE^2 = a^T\Sigma a$ gross = Σ|w_i|; net = Σw_i Underweight is not necessarily short. Net neutrality does not guarantee beta neutrality or low gross exposure.

ML2-D1 / MACHINE LEARNING / 21 QUESTIONS

Reinforcement Learning Problem Framing

Core understanding

Define the decision loop. The agent is the decision-maker; the environment applies the consequences of its actions. A state summarizes the information available when choosing, an action is a controllable choice, and a reward is feedback about the resulting transition. A trading environment can include market observations, execution, costs, and portfolio accounting; it need not consist only of the external market.

Why decisions are sequential. An action can change holdings, battery, queues, or time remaining, which changes the consequences and feasibility of later actions. A locally attractive gross return or shortest route need not maximize cumulative net reward. The relevant state therefore often includes information generated by the agent's own past choices.

Prediction and control answer different questions. Supervised learning fits supplied targets such as return or sentiment labels. RL estimates the consequences of actions from reward-bearing transitions and uses them to choose behavior. A supervised forecast may supply an RL state feature, and logged transitions can support policy learning; neither calling a label BUY nor putting rows in time order makes a supervised task RL.

Episodes, credit, and policy. An episode has modeled start and stopping rules; reaching a goal, forced liquidation, or a time cap can end a run. A continuing task can still be analyzed in daily chunks. Delayed rewards create credit-assignment problems because several earlier choices may contribute. A policy is the action-selection rule, deterministic or stochastic; a Q-table supplies estimates on which different policies may act.

Choose the formulation for the objective. RL is useful when actions have meaningful cumulative consequences and relevant experience can be obtained. Simulator fidelity, interaction cost, coverage, and safe deployment determine whether that formulation is practical. A fixed-label sentiment task is usually supervised classification; sequential rebalancing with turnover is a natural control problem. In reversed items, judge the claim first, then invert the marked True/False selection.

Every source group

G1 | Agent, Environment, State, Action, and Reward

ML2-D1G1Q1 p. 428 / ML2-D1G1Q2 p. 429 / ML2-D1G1Q3 p. 430

Classify each quantity by its role: observation/state, chosen action, or resulting reward. Include portfolio accounting and execution rules in the environment when they determine outcomes. Use a reward that reflects the stated goal, such as net value change or negative implementation shortfall.

Watch the wording: A useful market observation is not automatically a reward. The person who defines the task is not necessarily the agent acting in it.

G2 | Sequential Decision-Making

ML2-D1G2Q1 p. 430 / ML2-D1G2Q2 p. 431 / ML2-D1G2Q3 p. 432

Evaluate choices through the states and opportunities they create later. Include current inventory or resource capacity when it affects future costs and feasible actions. Immediate gross gain can conflict with cumulative net gain or successful completion.

Watch the wording: Independent daily optimization can overlook turnover and inventory accumulation. A fixed number of decisions does not make their order irrelevant.

G3 | Reinforcement Learning vs. Supervised Learning

ML2-D1G3Q1 p. 432 / ML2-D1G3Q2 p. 433 / ML2-D1G3Q3 p. 434

A supervised learner consumes supplied targets; RL consumes reward-bearing state-action transitions. Shared input indicators do not make objectives, targets, or outputs equivalent. Combine supervised perception or forecasts with a policy when each serves a distinct purpose.

Watch the wording: Renaming labels BUY and SELL does not create reinforcement learning. High classification accuracy does not guarantee the highest net policy return.

G4 | Episodes and Continuing Tasks

ML2-D1G4Q1 p. 434 / ML2-D1G4Q2 p. 435 / ML2-D1G4Q3 p. 436

Define initialization, termination, truncation, and any forced liquidation explicitly. A reporting boundary does not by itself reset a continuing task. Changing reset frequency can change trading costs and delayed consequences.

Watch the wording: The end of one episode is not the end of all training. Treating an artificial time cap as a genuine terminal event can change value targets; specify which task is modeled.

G5 | Delayed Rewards and Credit Assignment

ML2-D1G5Q1 p. 436 / ML2-D1G5Q2 p. 437 / ML2-D1G5Q3 p. 438

Connect a delayed outcome to all earlier actions that contributed to it. Zero rewards before a rare terminal payoff provide sparse informative feedback. Evaluate shaping rewards for loops, shortcuts, and changes to

the intended objective.

Watch the wording: The most recent action need not deserve all credit or blame. Paying for activity alone can reward motion or turnover instead of progress.

G6 | Policies and Action Selection

ML2-D1G6Q1 p. 438 / ML2-D1G6Q2 p. 439 / ML2-D1G6Q3 p. 440

A policy maps states to actions or action probabilities. Value estimates and the decision rule that uses them are distinct. Deterministic tie-breaking and fixed values are needed for reproducible greedy actions.

Watch the wording: A handwritten rule is still a policy. A deterministic environment can be controlled by a stochastic policy.

G7 | When Reinforcement Learning Is Appropriate

ML2-D1G7Q1 p. 440 / ML2-D1G7Q2 p. 441 / ML2-D1G7Q3 p. 442

Use sequential control when present choices alter future rewards, constraints, or decision context. Use supervised learning when the immediate task is reproducing supplied targets. Weigh the expected benefit of RL against the cost of collecting, validating, and deploying experience.

Watch the wording: A time series is not automatically an RL task. Recognizing a sequential problem does not make unrestricted live exploration appropriate.

Formulas and conditions

RL interaction

$s_t \to a_t \sim \pi(\cdot|s_t) \to (r_{t+1}, s_{t+1})$

The action is selected from current information; reward and successor state arrive after the transition.

Cumulative objective

$G_t = r_{t+1} + \gamma r_{t+2} + \gamma^2 r_{t+3} + \ldots$

Specify the horizon, reward definition, and reset rules before comparing policies.

Greedy policy

$\pi(s) \in \operatorname{argmax}_a Q(s,a)$

Argmax returns an action; max returns its value. Ties require a stated rule.

Work it through

Map an execution task

ML2-D1G1Q3 / pool p. 430

A parent order has 800 shares and 4 minutes left; the spread is 2 cents. The algorithm submits 200 shares and later observes $6 of implementation shortfall.

Remaining quantity, time, and spread are state inputs. Submitting 200 shares is the action. Execution and accounting are environment dynamics; a reward of -$6 penalizes the shortfall.

Result: The mapping determines what information and objective the controller uses.

Delayed feedback can reverse a local ranking

ML2-D1G5Q3 / pool p. 438

Suppose a held position earns rewards -2, +4, +4 over three steps; staying in cash earns zero. Use $\gamma = 0.9$.

The held position's return is $-2 + 0.9\times4 + 0.9^2\times4 = 4.84$. Its first reward is negative, but the complete discounted sequence is positive.

Result: An early cost does not by itself make a sequential decision poor.

ML2-D2 / MACHINE LEARNING / 21 QUESTIONS

Markov Decision Processes, Policies, and Values

Core understanding

A state must summarize relevant history. The Markov property says that the conditional distribution of the next state and reward depends on the current state and action, without needing extra past information. It is a sufficiency requirement, not a requirement to store every observation. Omitting package status, pending obligations, or holdings can merge contexts whose actions have different consequences.

Transitions describe uncertainty. For a fixed state-action pair, transition probabilities distribute mass across possible successor states and sum to one. Deterministic dynamics put all mass on one outcome. A deterministic policy cannot remove stochastic environment outcomes; an observed result also does not establish causation or change a probability to one.

Reward, return, and discounting. Reward is one transition's feedback; return aggregates future rewards with an explicit time convention. The first reward has weight one, then γ, γ^2, and so on. A larger γ makes distant outcomes relatively more influential, including delayed losses. Finite tasks with finite rewards can use $\gamma = 1$; continuing undiscounted sums require additional care.

Values depend on future behavior. $V^\pi(s)$ is expected return from a state under policy π. $Q^\pi(s,a)$ is expected return after choosing a particular action and then following π. V^π averages the policy's action values, whereas a greedy rule chooses a maximizing action. Expectations summarize distributions; they are neither guaranteed outcomes nor transition probabilities.

Bellman reasoning is a consistency relation. Compare immediate reward plus discounted expected continuation value, including every possible successor weighted by its probability. Greedy improvement is supported when the values and assumptions are sound. Choosing the largest incomplete or biased estimate is not a certificate of global optimality.

Every source group

G1 | States and the Markov Property

ML2-D2G1Q1 p. 444 / ML2-D2G1Q2 p. 445 / ML2-D2G1Q3 p. 446

A Markov state summarizes the information needed for relevant next-state and reward distributions. Include omitted context when it changes action consequences or constraints. A compact sufficient summary can replace raw history.

Watch the wording: The most important variable need not be a sufficient state. Past information is not irrelevant merely because it occurred earlier.

G2 | State Transitions and Environment Dynamics

ML2-D2G2Q1 p. 446 / ML2-D2G2Q2 p. 447 / ML2-D2G2Q3 p. 448

Condition transition probabilities on both state and action. Sum probabilities over all possible successors to one. Separate random market outcomes from deterministic changes to the agent's holdings.

Watch the wording: Stochasticity does not imply the agent has no influence. One realized successor does not prove a deterministic causal effect.

G3 | Rewards, Returns, and Objectives

ML2-D2G3Q1 p. 448 / ML2-D2G3Q2 p. 449 / ML2-D2G3Q3 p. 450

Distinguish an immediate reward from the return over a sequence. Use the specified numerical reward to determine the optimized objective. Include costs and relevant risk consequences when the stated objective requires them.

Watch the wording: Penalties are rewards with negative sign. Convergence cannot repair an incorrectly specified objective.

G4 | Discount Factor and Time Preference

ML2-D2G4Q1 p. 450 / ML2-D2G4Q2 p. 451 / ML2-D2G4Q3 p. 452

Weight the first reward by one and each later reward by an additional factor of γ. Low γ emphasizes near-term outcomes; high γ emphasizes delayed gains and losses. Finite-horizon undiscounted sums can be valid.

Watch the wording: γ does not rescale the first reward. A larger γ is not a promise of higher realized profit.

G5 | State-Value Functions

ML2-D2G5Q1 p. 452 / ML2-D2G5Q2 p. 453 / ML2-D2G5Q3 p. 454

Compute state value as expected return under the specified policy. Weight every outcome, including less likely losses. Allow different policies to assign different values to the same state.

Watch the wording: The most likely payoff is not generally the expected value. Zero current reward need not mean zero future value.

G6 | Action-Value Functions

ML2-D2G6Q1 p. 454 / ML2-D2G6Q2 p. 455 / ML2-D2G6Q3 p. 455

A Q-value fixes the first action and includes specified continuation behavior. For a stochastic policy, average Q-values using action probabilities. For greedy selection, compare all candidate Q-values

within the same state.

Watch the wording: Q-values do not need to sum to one. A negative expectation does not imply every realized trade loses.

G7 | Bellman Reasoning and Optimal Policies

ML2-D2G7Q1 p. 456 / ML2-D2G7Q2 p. 457 / ML2-D2G7Q3 p. 457

Add immediate reward to discounted expected successor value. A local policy-improvement step chooses a better-valued action under the available estimates. Optimality claims require appropriate dynamics, rewards, coverage, and value accuracy.

Watch the wording: Do not compare successor values while ignoring different immediate rewards. One improved state does not prove the entire policy is optimal.

Formulas and conditions

Markov sufficiency

$P(s_{t+1}, r_{t+1} \mid \text{history}, a_t) = P(s_{t+1}, r_{t+1} \mid s_t, a_t)$

Equality is a modeling assumption about the chosen state representation.

Discounted return

$G_t = \Sigma_{k=0}^{H-1} \gamma^k r_{t+k+1}$

For finite H and finite rewards, $\gamma = 1$ is permitted. For bounded continuing rewards, $\gamma < 1$ gives an absolutely convergent sum.

State and action values

$V^\pi(s) = \Sigma_a \pi(a|s) Q^\pi(s,a)$; $Q^\pi(s,a) = E[r + \gamma V^\pi(s') \mid s,a]$

The same continuation policy π must be used in both quantities.

Bellman optimality

$V^*(s) = \max_a E[r + \gamma V^*(s') \mid s,a]$

The expectation accounts for stochastic transitions; a true terminal state has no future reward.

Work it through

Expected value is not the most likely payoff

ML2-D2G5Q1 / pool p. 452

State A yields return 10 with probability 0.8 and -10 with probability 0.2; State B yields 5 for certain.

$V(A) = 0.8 \times 10 + 0.2 \times (-10) = 6$; $V(B) = 5$. An expected-return objective ranks A higher despite its downside outcome.

Result: A is worth 6 in expectation, not a guaranteed 10.

One-step Bellman comparison

ML2-D2G7Q1 / pool p. 456

Safe gives 2 now then value 5. Risky gives 0 now then value 10 or -2 with equal probability. Let $\gamma = 0.9$.

Safe: $2 + 0.9 \times 5 = 6.5$. Risky successor expectation: $0.5 \times 10 + 0.5 \times (-2) = 4$; total: $0 + 0.9 \times 4 = 3.6$.

Result: Choose Safe under these estimates.

Average over the policy

ML2-D2G6Q2 / pool p. 455

A policy selects A with probability 0.75 and B with probability 0.25; $Q^\pi(s,A)=8$ and $Q^\pi(s,B)=0$.

$V^\pi(s) = 0.75 \times 8 + 0.25 \times 0 = 6$. Using max would answer a different action-selection question.

Result: The current policy's state value is 6.

ML2-D3 / MACHINE LEARNING / 21 QUESTIONS

Q-Learning Mechanics

Core understanding

Read a Q-table correctly. Each row identifies a represented state and each column an available action. Entries estimate cumulative action values, not immediate rewards, probabilities, or exact successors. Greedy selection compares a row; ties are resolved by an implementation rule. An unvisited row still reflects initialization, even if its entries look numerically plausible.

The update blends old knowledge with a target. For observed (s,a,r,s′), standard Q-learning forms y = r + γ max_a′ Q(s′,a′). It then changes only Q(s,a) by α[y - Q(s,a)]. The TD error may be negative even when the successor value is positive. The target is one sample using current estimates; repeated representative transitions are needed in stochastic settings.

α, γ, and exploration have different jobs. The learning rate α controls how strongly a new target changes the old estimate: zero freezes it, one replaces it. The discount factor γ controls the weight of future rewards inside the target. Epsilon governs behavioral exploration. A high constant α reacts quickly but remains noisy; a declining schedule supports stationary estimation, while a positive rate can allow adaptation to change.

Off-policy bootstrapping. Q-learning bootstraps from the maximum next-state estimate even when the behavior policy next chooses a different exploratory action. That separates the behavior policy collecting data from the greedy target policy. A true terminal transition uses target r alone; it must not bootstrap from the reset state of the next episode. Max over noisy estimates can propagate overestimation.

Convergence needs more than stability. For tabular stationary finite problems, sufficient repeated visitation and suitable learning-rate conditions support convergence results. A flat curve, unchanged actions, optimistic initialization, or one visit per pair is not a proof. Sparse coverage, hidden state, nonstationarity, and biased simulations weaken any conclusion drawn from small numerical updates.

Every source group

G1 | Q-Table Structure and Interpretation

ML2-D3G1Q1 p. 460 / ML2-D3G1Q2 p. 461 / ML2-D3G1Q3 p. 462

Use one value per state-action pair and compare actions within the current row. Q estimates expected cumulative return under the represented continuation behavior. State the tie-breaking rule when extracting a greedy policy.

Watch the wording: Negative Q can still be the largest value in a row. A high value in one row does not rank that action in every state.

G2 | Temporal-Difference Update

ML2-D3G2Q1 p. 462 / ML2-D3G2Q2 p. 463 / ML2-D3G2Q3 p. 463

Compute the target, then the TD error, then the weighted update. Update the previous state-action pair using the observed reward and successor. Repeated transitions can revise the same entry without an explicit transition model.

Watch the wording: A negative reward need not replace the entire estimate. One actual sample is not an exact expected value.

G3 | Learning Rate

ML2-D3G3Q1 p. 464 / ML2-D3G3Q2 p. 465 / ML2-D3G3Q3 p. 465

α weights new targets against the existing estimate. α = 0 freezes; α = 1 adopts the sample target. Large constant α emphasizes recent outcomes and can retain noise; smaller α adapts more slowly.

Watch the wording: Constant α is not an equal-weight mean of all historical samples. No fixed learning rate guarantees the best final policy across tasks.

G4 | Discount Factor in Q-Learning

ML2-D3G4Q1 p. 466 / ML2-D3G4Q2 p. 466 / ML2-D3G4Q3 p. 467

γ weights the successor term while leaving immediate reward undiscounted. γ = 0 still leaves an α-weighted immediate-reward update. Changing γ can alter the preferred policy without changing the reward function.

Watch the wording: γ is not the random-action rate. γ = 1 does not imply divergence in every finite episodic task.

G5 | Maximum Future Q-Value

ML2-D3G5Q1 p. 468 / ML2-D3G5Q2 p. 468 / ML2-D3G5Q3 p. 469

Use the maximum available successor estimate in standard Q-learning. For a true terminal transition, set future contribution to zero. Noisy inflated maxima can contaminate predecessor estimates through bootstrapping.

Watch the wording: The selected exploratory next action need not supply the Q-learning target. The maximum estimate is not proof of the maximum true value.

G6 | Off-Policy Learning

ML2-D3G6Q1 p. 469 / ML2-D3G6Q2 p. 470 / ML2-D3G6Q3 p. 471

Separate the behavior that samples experience from the policy represented in the target. Exploratory actions can provide useful evidence about good and bad alternatives. A max-successor target differs from bootstrapping on the actually selected next action.

Watch the wording: Off-policy does not mean learning without adequate state-action coverage. Using a Q-table does not by itself make an algorithm on-policy.

G7 | Initialization, Visitation, and Convergence

ML2-D3G7Q1 p. 471 / ML2-D3G7Q2 p. 472 / ML2-D3G7Q3 p. 473

Inspect visitation counts alongside Q-values and learned actions. Initialization can strongly influence early greedy behavior. Require sustained representative coverage and suitable update conditions before making convergence claims.

Watch the wording: An unchanged unvisited row contains initialization, not learned evidence. Optimistic initialization does not guarantee exploration of every reachable pair.

Formulas and conditions

TD target and error

$y = r + \gamma \max_{a'} Q(s',a'); \delta = y - Q(s,a)$

Use $y = r$ for a genuinely terminal transition. A time-limit truncation needs the task's explicit continuation convention.

Q-learning update

$Q_{new}(s,a) = Q_{old}(s,a) + \alpha\delta = (1-\alpha)Q_{old}(s,a) + \alpha y$

Other table entries remain unchanged in a single direct tabular update; $0 \le \alpha \le 1$ is the usual blending range.

Learning-rate convergence conditions

$\sum_n \alpha_n(s,a) = \infty$ and $\sum_n \alpha_n(s,a)^2 < \infty$

These conditions apply per repeatedly visited pair for a finite stationary Markov task with bounded rewards and $\gamma < 1$; suitable episodic variants need their own termination assumptions. Merely decreasing α is not enough.

Greedy action

$a^* \in \operatorname{argmax}_a Q(s,a)$

Break ties explicitly. The value max Q is a number; argmax identifies an action.

Work it through

Positive TD update

ML2-D3G2Q1 / pool p. 462

Old Q = 2, reward = 1, $\gamma = 0.9$, max successor Q = 4, $\alpha = 0.5$.

Target $y = 1 + 0.9\times4 = 4.6$. Error $\delta = 4.6 - 2 = 2.6$. New $Q = 2 + 0.5\times2.6 = 3.3$.

Result: New Q is 3.3, halfway between 2 and 4.6.

Negative TD update

ML2-D3G2Q2 / pool p. 463

Old Q = 5, reward = -2, max successor Q = 3, $\gamma = 0.5$, $\alpha = 0.2$.

Target $= -2 + 0.5\times3 = -0.5$. Error $= -0.5 - 5 = -5.5$. New $Q = 5 + 0.2\times(-5.5) = 3.9$.

Result: A positive successor estimate does not prevent the current value from falling.

Terminal reward counted once

ML2-D3G5Q3 / pool p. 469

A goal transition pays 20 and terminates; old Q = 8, $\alpha = 0.25$, $\gamma = 0.9$.

No reward remains after termination, so $y = 20$ and $Q_{new} = 8 + 0.25\times(20 - 8) = 11$.

Result: New Q is 11; do not add the reward again or bootstrap from the next episode.

ML2-D4 / MACHINE LEARNING / 21 QUESTIONS

Exploration, Exploitation, and Learning Dynamics

Core understanding

Exploration buys information. Exploitation selects the currently highest estimated action. Exploration tries alternatives that may be worse now but can improve later decisions. Early lucky samples can make an inferior action look best, while continual heavy randomness can reduce realized performance even after values are useful. The right balance depends on how much future use the information will receive.

Read the exact epsilon convention. In the common implementation, the exploratory branch samples uniformly from all actions, so it can select the greedy action too. Other implementations sample only nongreedy actions. Ties in the greedy set add another source of randomness. Epsilon is a branch probability, not the marginal probability assigned to every action.

Decay does not certify coverage. A geometric schedule reduces random actions, but fast decay can freeze early mistakes. A positive floor preserves opportunities to sample alternatives, yet it cannot make unreachable states reachable or ensure equal visitation. State coverage and state-action coverage differ, and one sample per pair can miss stochastic variability or rare losses.

Measure learning and deployment separately. A flat training reward curve can reflect adequate learning, poor exploration, or a restrictive state representation. Compare seeds and held-out policies, with controlled tie-breaking and fixed preprocessing. Greedy evaluation can remove deliberate exploration; it does not make inaccurate values correct or eliminate environment randomness.

Distinguish noise from change. A stationary stochastic environment can produce different outcomes from the same pair while its distribution stays fixed. Nonstationarity changes the distribution itself, for example when trading costs rise. Frozen Q-values do not adapt; retraining or online updating can respond but introduces estimation noise and, with exploration, real costs.

Every source group

G1 | Exploration vs. Exploitation

ML2-D4G1Q1 p. 475 / ML2-D4G1Q2 p. 476 / ML2-D4G1Q3 p. 476

Exploitation follows the best current estimate; exploration acquires evidence about alternatives. Judge exploratory choices by possible information benefits as well as immediate reward. Too much exploration can depress deployment reward despite accurate values.

Watch the wording: The current leader may be an estimation error. One disappointing trial does not fully characterize a stochastic alternative.

G2 | Epsilon-Greedy Action Selection

ML2-D4G2Q1 p. 477 / ML2-D4G2Q2 p. 478 / ML2-D4G2Q3 p. 478

Inspect whether the exploratory set includes the greedy action. Add probability from every branch that can produce a given action. Account separately for ties among maximizing actions.

Watch the wording: ε is not the probability of each nongreedy action. $\varepsilon = 0$ removes exploration but not random greedy tie-breaking.

G3 | Exploration Decay

ML2-D4G3Q1 p. 479 / ML2-D4G3Q2 p. 479 / ML2-D4G3Q3 p. 480

For multiplicative decay, apply the decay factor once per stated step. Fast decay can leave useful alternatives untested. A positive floor preserves residual exploration, with a continuing short-term cost.

Watch the wording: Positive geometric decay does not reach exactly zero in finite exact-arithmetic steps. A floor does not guarantee equal or sufficient visitation.

G4 | State-Action Coverage

ML2-D4G4Q1 p. 481 / ML2-D4G4Q2 p. 481 / ML2-D4G4Q3 p. 482

Count state-action pairs, not just distinct states. Exploratory choices may unlock downstream states a greedy route never reaches. Use repeated samples when rewards and transitions are stochastic.

Watch the wording: Unvisited does not necessarily mean unreachable. One visit per pair is nominal coverage, not accurate estimation.

G5 | Premature Convergence and Local Behavior

ML2-D4G5Q1 p. 482 / ML2-D4G5Q2 p. 483 / ML2-D4G5Q3 p. 484

Consider early-sample luck when behavior stabilizes rapidly. Use multiple seeds and validation evidence to assess sensitivity. Check alternative actions and coverage before treating a flat curve as completion.

Watch the wording: The majority policy across seeds need not be the best. Longer training alone is not a global-optimality guarantee.

G6 | Training vs. Deployment Behavior

ML2-D4G6Q1 p. 484 / ML2-D4G6Q2 p. 485 / ML2-D4G6Q3 p. 485

Separate training exploration from the behavior being evaluated or deployed. Freeze values, preprocessing, and tie rules for a fixed-policy evaluation. Report variability if the evaluated policy intentionally

remains stochastic.

Watch the wording: The deployment epsilon need not equal the training epsilon. Greedy actions can still be poor when values are inaccurate.

G7 | Nonstationary and Stochastic Environments

ML2-D4G7Q1 p. 486 / ML2-D4G7Q2 p. 487 / ML2-D4G7Q3 p. 487

Different realized outcomes can occur under a fixed stochastic distribution. Changing transition or reward distributions can make historical values stale. Monitor drift and assess the costs of adaptation and residual exploration.

Watch the wording: Random reward variation alone does not prove regime change. Nonzero epsilon does not instantly detect every environmental change.

Formulas and conditions

Uniform exploration over all actions

P(unique greedy action) = 1-ε+ε/m; P(each nongreedy action) = ε/m

There are m actions and a unique greedy action. If k actions tie and greedy ties are uniform, each tied action gets (1-ε)/k + ε/m.

Exploration excludes the greedy action

P(greedy) = 1-ε; P(each nongreedy) = ε/(m-1)

Applies only to the stated nongreedy-only exploratory branch with one greedy action.

Geometric decay

ε_n = ε_0 d^n; with floor: ε_n = max(ε_min, ε_0 d^n)

Count decay events, not raw decisions unless the implementation decays after every decision; 0 < d < 1.

Tabular coverage

Total entries = |S|×|A|; pair coverage = visited pairs / total entries

Counts alone do not measure sample quality, rare-outcome coverage, or deployment reachability.

Work it through

Exploration can choose the greedy action

ML2-D4G2Q1 / pool p. 477

Three actions, ε = 0.20, a unique greedy action, and uniform exploratory sampling over all three.

Greedy probability = 0.80 + 0.20/3 = 0.8667. Each other action receives 0.20/3 = 0.0667. The probabilities sum to one.

Result: 86.67% greedy selection, despite an 80% greedy branch.

A different implementation gives different probabilities

ML2-D4G2Q2 / pool p. 478

Four actions, ε = 0.40, and exploration samples only the three nongreedy actions.

Greedy selection has probability 0.60. Each other action has probability 0.40/3 ≈ 0.1333.

Result: Always read the sampling set before calculating probabilities.

Decay versus evidence

ML2-D4G3Q2 / pool p. 479

Begin at ε = 0.80 and multiply by 0.90 after each episode. What is epsilon after three decays?

ε_3 = 0.80×0.90³ = 0.5832. The number describes the exploratory branch probability, not how many pairs have been adequately sampled.

Result: 58.32% after three decay events.

ML2-D5 / MACHINE LEARNING / 21 QUESTIONS

Dyna and Model-Based Reinforcement Learning

Core understanding

Dyna combines acting, modeling, and planning. Plain Q-learning can update from a real transition without learning an environment model. Dyna-Q adds a model of next states and rewards, then samples model-generated transitions for extra Q-updates. These simulated transitions reuse existing information; they are not new external observations or fresh realized profits.

Represent the dynamics you actually observe. A deterministic stationary transition can be stored as one successor and reward for each observed pair. Stochastic environments require a distribution or another representation of variability. Normalized successor counts estimate transition probabilities, and a reward model can estimate an average rather than the last noisy observation. Unknown pairs need explicit treatment instead of invented default transitions.

Planning spreads information through the table. After a real update and model update, each planning step selects a modeled pair, generates a successor and reward, and applies the usual TD rule. A newly discovered terminal reward can spread to predecessor values without physically traversing the entire route again. Which pairs are sampled and in what order affects propagation speed.

Sample efficiency is not free computation. Dyna can improve the policy using fewer costly real interactions, at the price of more computation per interaction. More planning is attractive when an accurate model is cheap relative to acting. If direct simulation is already inexpensive or model construction is inaccurate and costly, the practical benefit can disappear.

Planning repeats model assumptions, including mistakes. Extensive planning can amplify omitted costs, stale routes, optimistic tails, and missing rare outcomes. More replays of a biased model do not create corrective evidence. Continued direct observation, targeted coverage, and model validation remain necessary, especially after environmental change.

Every source group

G1 | Direct vs. Simulated Experience

ML2-D5G1Q1 p. 490 / ML2-D5G1Q2 p. 491 / ML2-D5G1Q3 p. 492

Direct experience comes from the external environment; planning generates transitions internally. Both types can drive the same Q-learning update. Planning can revisit an older modeled pair instead of only the most recent real pair.

Watch the wording: Fifty planning steps are not fifty independent market observations. Model-free direct Q-learning does not require a learned transition table.

G2 | Transition Model Learning

ML2-D5G2Q1 p. 492 / ML2-D5G2Q2 p. 493 / ML2-D5G2Q3 p. 493

Store deterministic successors only when that representation matches the dynamics. Use outcome counts or probabilities when one pair has multiple possible successors. Restrict empirical replay to supported pairs or make any assumptions about unobserved pairs explicit.

Watch the wording: The latest observed outcome is not the full stochastic distribution. Replaying an old route does not prove it is still open.

G3 | Reward Model Learning

ML2-D5G3Q1 p. 494 / ML2-D5G3Q2 p. 495 / ML2-D5G3Q3 p. 495

A reward model supplies the immediate reward estimate for a simulated target. Average representative noisy rewards when expected reward is the desired model. Revise the model if costs or reward behavior change.

Watch the wording: An accurate transition model does not imply an accurate reward model. A modeled reward is not a fresh external payoff.

G4 | Dyna Planning Updates

ML2-D5G4Q1 p. 496 / ML2-D5G4Q2 p. 496 / ML2-D5G4Q3 p. 497

Perform a direct Q-update, update the model, then run the configured planning updates. Use the same α and γ logic in simulated TD updates. Relevant predecessor replay can propagate a discovered reward backward.

Watch the wording: dyna = 0 leaves ordinary direct Q-learning in place. One replay does not instantly solve every upstream state.

G5 | Sample Efficiency

ML2-D5G5Q1 p. 498 / ML2-D5G5Q2 p. 498 / ML2-D5G5Q3 p. 499

Measure real interactions needed to reach a learning target separately from compute cost. Planning can reuse scarce or expensive direct observations. Compare model cost and fidelity against the cost of obtaining more direct experience.

Watch the wording: More internal updates do not create independent information. Model-based learning does not guarantee superiority on every run.

G6 | Model Error and Planning Bias

ML2-D5G6Q1 p. 499 / ML2-D5G6Q2 p. 500 / ML2-D5G6Q3 p. 501

Planning propagates the transition and reward errors already present in the model. Fresh direct observations are needed to discover unmodeled changes. Reducing planning can limit amplification but does not correct the underlying bias.

Watch the wording: Thousands of optimistic replays do not prove positive true expected return. Nonstationarity can invalidate both the policy and its internal model.

G7 | Dyna in Deterministic and Stochastic Environments

ML2-D5G7Q1 p. 501 / ML2-D5G7Q2 p. 502 / ML2-D5G7Q3 p. 502

A deterministic table can exactly represent observed stationary deterministic pairs. For stochastic outcomes, estimate and sample the categorical successor distribution. Continue collecting evidence about rare outcomes and previously unvisited pairs.

Watch the wording: No observed rare penalty does not imply zero probability. A sampled next state is an outcome, not necessarily the expected next-state representation.

Formulas and conditions

Transition model from counts

$\hat{T}(s'|s,a) = N(s,a,s') / \Sigma_u N(s,a,u)$

The denominator must be positive. Lecture pseudocounts avoid division by zero but encode a prior, not evidence of observed outcomes.

Reward-model update

$\hat{R}_{new}(s,a) = (1-\beta)\hat{R}_{old}(s,a) + \beta r$

The lecture uses a learning-rate blend. Constant β emphasizes recent rewards; β = 1/n gives a running mean over n observations for a fixed pair.

Dyna computation per real transition

Q-updates per real step = 1 + d; total = $N_{real}(1+d)$

Assumes exactly one direct update and d planning updates per real transition. Model updates are additional bookkeeping.

Planning target

$y_{model} = \hat{r} + \gamma \max_{a'} Q(s'_{model},a')$

Generate the modeled successor and reward from the selected pair. True terminal modeled transitions omit the future term.

Work it through

Count updates without inventing data

ML2-D5G4Q1 / pool p. 496

An agent gets 200 real transitions and performs dyna = 20 planning steps after each.

Direct Q-updates = 200. Planning updates = 200×20 = 4,000. Total Q-updates = 4,200, all based on only 200 real transitions.

Result: More updates can improve sample efficiency but do not create 4,200 independent observations.

Estimate a stochastic model

ML2-D5G7Q2 / pool p. 502

From one pair, a model records 50 up, 20 flat, and 30 down outcomes.

Total count = 100; estimated probabilities are 0.50, 0.20, and 0.30. Planning may sample any of the three outcomes using those probabilities.

Result: A single last-outcome record would discard the estimated distribution.

A rare event can be missing by chance

ML2-D5G7Q3 / pool p. 502

An action has an independent 1% severe-penalty probability per trial. No penalty appears in the first 50 trials.

P(no penalty in 50 trials) = 0.99^50 ≈ 0.605. Missing the event is therefore quite plausible. Replaying only the observed safe outcome cannot reveal it.

Result: Absence in 50 samples is weak evidence of safety, not proof that the true probability is zero.

ML2-D6 / MACHINE LEARNING / 21 QUESTIONS

State Representation, Discretization, and Reward Design

Core understanding

Choose sufficient, available information. State features should help distinguish consequences, constraints, or valuable actions and must be available at decision time. Momentum, volatility, event status, inventory, and remaining time can each matter. Adding random identifiers or many redundant indicators increases complexity without necessarily improving decisions; adding tomorrow's outcome creates leakage.

Discretization trades detail for evidence per state. Binning maps continuous observations into a finite set of labels. Fixed thresholds retain fixed numeric meaning but can become imbalanced after a distribution shift. Rank-based thresholds spread reference observations across bins but change meaning with the reference distribution. Fit thresholds using permitted historical data, define boundary behavior, and apply the intended deployment convention consistently.

The Cartesian product grows quickly. With b bins for each of k indicators, the nominal market-state space is b^k. Holdings and other dimensions multiply it further, and each action needs a separate value. Fine bins can separate meaningful contexts but produce sparse visitation; coarse bins pool experience but can merge contexts needing different actions. Not every nominal combination must be reachable.

Aliasing is missing decision information. Two observations share a state label when the learner cannot distinguish them. That is harmful when their dynamics, rewards, or feasible actions differ. Inventory direction alone may be insufficient if position magnitude changes risk limits. Adding relevant context or a useful history summary can help approximate Markov sufficiency, but neither a larger table nor a higher learning rate recovers information absent from the state.

Rewards specify incentives, not aspirations. Realized-only P&L delays feedback while open positions can gain or lose economic value. Mark-to-market net value changes provide denser feedback but still need appropriate risk objectives and timing. Penalties and bonuses can change the optimal policy: extreme turnover penalties may cause inactivity, and flat-position bonuses may encourage costly liquidation-and-reentry cycles. Audit full behavior rather than one rewarded metric.

Every source group

G1 | State Feature Selection

ML2-D6G1Q1 p. 505 / ML2-D6G1Q2 p. 506 / ML2-D6G1Q3 p. 507

Keep features available at decision time and relevant to future actions, outcomes, or constraints. Treat current holdings and time remaining as legitimate state features. Balance added information against redundancy, sparse visitation, and limited data.

Watch the wording: Training-file availability does not imply live availability. A univariate historical correlation does not guarantee policy value.

G2 | Continuous-to-Discrete State Mapping

ML2-D6G2Q1 p. 507 / ML2-D6G2Q2 p. 508 / ML2-D6G2Q3 p. 509

Define numeric thresholds and inclusion at each boundary. Expect multiple raw observations to map to the same finite label. Fit threshold statistics on allowed reference data and consider boundary noise.

Watch the wording: Bins do not preserve exact distances or all economic distinctions. Balanced reference counts do not guarantee equal deployment counts or an optimal state.

G3 | Granularity and State-Space Size

ML2-D6G3Q1 p. 509 / ML2-D6G3Q2 p. 510 / ML2-D6G3Q3 p. 511

Multiply bin counts across dimensions to get nominal combinations. Multiply by action count to size a Q-table. Refine decision-relevant dimensions selectively when broad refinement would exhaust coverage.

Watch the wording: Doubling every dimension's bins multiplies states exponentially in dimension count. Larger tables do not supply their own training observations.

G4 | State Aliasing and Missing Information

ML2-D6G4Q1 p. 511 / ML2-D6G4Q2 p. 512 / ML2-D6G4Q3 p. 513

Look for identical encoded states with materially different action consequences. Add available context or history summaries that distinguish those cases. Recognize that any summary can still omit relevant hidden information.

Watch the wording: Perfect measurement of included variables does not make an incomplete state sufficient. Raising α cannot reveal a missing direction, event flag, or hidden condition.

G5 | Position and Holdings as State

ML2-D6G5Q1 p. 513 / ML2-D6G5Q2 p. 514 / ML2-D6G5Q3 p. 514

Include current position when order size, cost, or feasibility depends on holdings. Represent position magnitude or capacity if sign alone merges different constraint states. Check executed holdings and enforce hard limits independently of the learned policy.

Watch the wording: Agent-controlled information can belong in state. Adding holdings does not guarantee that the learner stops overtrading.

G6 | Reward Definition

ML2-D6G6Q1 p. 515 / ML2-D6G6Q2 p. 516 / ML2-D6G6Q3 p. 516

Distinguish realized P&L from total economic value change. Incorporate commissions and impact when optimizing net results. Align reward scale, timing, discounting, and risk treatment with the long-run objective.

Watch the wording: Zero realized profit does not mean zero economic change. Cumulative learning cannot turn an arbitrary reward into risk-adjusted growth automatically.

G7 | Reward Shaping and Unintended Incentives

ML2-D6G7Q1 p. 517 / ML2-D6G7Q2 p. 518 / ML2-D6G7Q3 p. 518

Evaluate shaping terms for changes in incentives and complete portfolio outcomes. Calibrate penalties to actual costs or an explicitly intended preference. Check for activity loops, excessive inactivity, and liquidation/reentry behavior.

Watch the wording: Reward maximization can succeed while the human objective fails. Arbitrary shaping is not guaranteed to preserve the optimal policy.

Formulas and conditions

Nominal state-space size

$|S| = h \times \prod_{j=1}^{k} b_j$; $|Q| = |S| \times |A|$

b_j is the bin count of feature j, h is the number of holdings states; nominal combinations can include unreachable states.

Mixed-radix state index

$s = (i_1 \times B_2 + i_2) \times B_3 + i_3$

Three-feature example: i_j is the zero-based bin index and B_j its bin count. Extend by nesting for more dimensions. Equal ten-bin dimensions can be encoded like decimal digits.

Daily economic reward

$r_{t+1} = V_{t+1}^{net} - V_t^{net}$

Specify the valuation and execution times; include realized and unrealized P&L, deduct costs once, and adjust for external cash flows if present.

Illustrative shaped reward

$r_{shaped} = r_{net} - \lambda \times turnover + bonus(s,a,s')$

This is a design choice and can change the optimal policy; λ and bonuses require economic validation.

Work it through

Doubling bins multiplies complexity

ML2-D6G3Q1 / pool p. 509

Four indicators move from 5 bins each to 10 bins each; the same history is used.

Old market states = $5^4 = 625$. New states = $10^4 = 10{,}000$. Ratio = 16. With 3 holdings states and 3 actions, entries rise from 5,625 to 90,000.

Result: Sixteen times as many entries require much more evidence for comparable coverage.

The same signal can imply different orders

ML2-D6G5Q2 / pool p. 514

For target +1,000 shares, compare portfolios currently at 0, +1,000, and -1,000 shares.

Required orders are +1,000, 0, and +2,000 shares. The market signal can be identical while costs and feasibility differ.

Result: Holdings distinguish entering, maintaining, and reversing a position.

A reward loop that misses the task

ML2-D6G7Q2 / pool p. 518

A robot earns +1 for each forward move toward the goal and 0 for moving backward; it alternates forward and backward without finishing.

Each two-action cycle earns +1 despite zero net progress. Ten cycles earn +10 while completion never occurs. A symmetric net-progress measure would make the cycle's net progress zero.

Result: The agent exploits the numerical reward; completion and full-path behavior must be checked.

ML2-D7 / MACHINE LEARNING / 21 QUESTIONS

Reinforcement Learning for Trading

Core understanding

Market state must respect the clock. A daily trading state can combine discretized indicators with current holdings and constraints. Use only information available when the action is selected, including the permitted publication time of external data. Full-history normalization, future returns in indicators, and test-based bin selection contaminate a claimed out-of-sample evaluation.

An action is not necessarily an order. If actions specify target positions, order quantity equals desired holdings minus current holdings. Choosing long while already long can require no trade; switching short to long crosses zero and trades twice the position magnitude. Order clipping or rejection can leave actual holdings different from requested targets, so the environment must report the realized state.

Positions persist and generate rewards. Holding an asset can change portfolio value even on days with no orders. A reward based only on position closure delays economic feedback; daily net value changes include unrealized P&L. The action comes first, then its interval's outcome is observed. Future outcomes may be realized rewards later, but cannot be supplied as earlier decision features.

Costs change the decision problem. Commissions and impact affect the value of trading and therefore can change Q-value rankings. Frequent reversals can erase a small gross advantage. Training on gross rewards and evaluating only net results can produce a systematically misaligned policy; cost-aware training can rationally suppress marginal trades without implying that the learner failed.

Evaluate the policy that will actually run. Fit features, thresholds, rewards, and hyperparameters on training and development periods. For a frozen-policy test, keep values and preprocessing fixed and use an untouched later period. Repeated tuning on a test makes it development data. Monitor later state frequencies, costs, and outcomes: stable actions do not imply stable profitability, and neither rolling retraining nor online adaptation guarantees improvement.

Every source group

G1 | Mapping Market Data into States

ML2-D7G1Q1 p. 521 / ML2-D7G1Q2 p. 522 / ML2-D7G1Q3 p. 523

Build state from available indicators, portfolio information, and constraints. Fit preprocessing and choose bins without using held-out outcomes. Check coverage and aliasing after discretization.

Watch the wording: A historical-price formula can still leak if its window includes future observations. Sharing a bin does not establish identical future-return distributions.

G2 | Mapping Actions into Trades or Positions

ML2-D7G2Q1 p. 523 / ML2-D7G2Q2 p. 524 / ML2-D7G2Q3 p. 525

Specify whether an action is an incremental order or a desired target. Translate targets using current executed holdings. Apply feasibility constraints and pass actual execution outcomes back into state.

Watch the wording: HOLD need not mean cash. A target action does not remove commission, impact, or liquidity constraints.

G3 | Holdings, Inventory, and Position Persistence

ML2-D7G3Q1 p. 525 / ML2-D7G3Q2 p. 526 / ML2-D7G3Q3 p. 527

Allow positions to persist across decisions without repeated orders. Include holdings detail sufficient for costs, risk, and capacity. Account for P&L on held positions even when trade quantity is zero.

Watch the wording: No order is not the same as no exposure or zero reward. Long/cash/short labels can alias very different position sizes.

G4 | Trading Rewards and Portfolio Value

ML2-D7G4Q1 p. 527 / ML2-D7G4Q2 p. 528 / ML2-D7G4Q3 p. 528

Reward timing must follow the economic interval being measured. Use net total portfolio value when the objective includes open-position gains, losses, and costs. Distinguish dollar reward from percentage return using starting value.

Watch the wording: A realized-only zero reward can hide an unrealized loss. Paying for order count rewards activity even if the portfolio loses money.

G5 | Transaction Costs, Impact, and Turnover

ML2-D7G5Q1 p. 529 / ML2-D7G5Q2 p. 530 / ML2-D7G5Q3 p. 530

Measure turnover from the actual position changes required by targets. Train with relevant cost consequences when seeking net performance. Recognize that removing marginal trades can be a rational response to costs.

Watch the wording: A reversal trades twice the one-sided position magnitude. A positive gross opportunity need not remain positive after costs.

G6 | Training, Testing, and Out-of-Sample Evaluation

ML2-D7G6Q1 p. 531 / ML2-D7G6Q2 p. 531 / ML2-D7G6Q3 p. 532

Use chronological training, selection, and untouched evaluation periods. Freeze the policy and fitted preprocessing for a fixed-policy test. Compare

in-sample and out-of-sample performance with costs and benchmarks.

Watch the wording: Selecting among many runs on the test return overfits that test. One poor test does not prove every RL trading design is useless.

G7 | Market Nonstationarity and Policy Stability

ML2-D7G7Q1 p. 533 / ML2-D7G7Q2 p. 533 / ML2-D7G7Q3 p. 534

Compare recent state-action outcomes with historical estimates. Monitor distribution, costs, turnover, performance, and realized holdings. Validate retraining or online adaptation as its own deployment process.

Watch the wording: Unchanged actions do not prove unchanged market rewards. Newer data and continuing exploration do not automatically improve performance.

Formulas and conditions

Target-to-order mapping

$\Delta H_t = H_{target,t} - H_{current,t}$

Use executed holdings, not a previous requested target; clipping or partial fills can create a difference.

Net dollar reward and return

$r_t = V_{end,net} - V_{start}$; $R_t = r_t / V_{start}$

Assumes no external cash flow and positive starting value. Do not subtract a commission twice if already included in net value.

Simple explicit cost model

$cost_t = c \times 1[\Delta H_t \neq 0] + \eta P_t |\Delta H_t|$

Illustrative per-order commission c plus proportional impact η; actual question-specific execution conventions take precedence.

Three-indicator market states

$3 \times 3 \times 3 = 27$; with 3 holdings categories: 81 states

Assumes all Cartesian combinations are encoded; count visitation separately.

Work it through

A reversal crosses through cash

ML2-D7G2Q1 / pool p. 523

Current holdings are -1,000 shares and the action requests a +1,000-share target.

ΔH = +1,000 - (-1,000) = +2,000 shares. By comparison, cash-to-long buys 1,000 and long-to-long buys zero.

Result: Buy 2,000 shares, with costs based on the actual reversal.

Net reward in dollars and percent

ML2-D7G4Q2 / pool p. 528

Start at $100,000. End value before a $50 commission is $101,000.

Net ending value = $101,000 - $50 = $100,950. Net reward = $950. Net return = $950/$100,000 = 0.0095 = 0.95%.

Result: The return is 0.95%, not 9.5%.

Cost-aware opportunity threshold

ML2-D7G5Q2 / pool p. 530

Suppose a reversal trades 2,000 shares at $50; commission is $10 and modeled impact is 0.1% of traded notional. Expected gross gain is $80.

Notional = 2,000×$50 = $100,000. Impact = $100. Total cost = $110. Expected net gain = $80 - $110 = -$30.

Result: A gross-positive trade can be unattractive once execution costs enter the value.

ML2-D8 / MACHINE LEARNING / 21 QUESTIONS

Comparative Strategy Learning and Evaluation

Core understanding

Compare complete systems, not output labels. A tree strategy fits supplied labels or returns and then maps predictions into trades. A Q-learner estimates action values from reward-bearing transitions and extracts a policy. LONG is a decision, not necessarily a return forecast. The two can use the same indicators while differing in objectives, horizons, feedback, and portfolio-state access.

The forecast-to-trade step has its own economics. A return model and its decision threshold are distinct components. Changing a threshold can change positions without refitting the predictor, and a holdings-aware no-trade region can account for costs. Direct policy learning can encode these consequences in rewards, but neither architecture automatically aligns with net risk-adjusted performance.

Learning signals embed assumptions. Supervised labels specify a horizon, thresholds, and possibly costs or an oracle position rule. They legitimately use later outcomes within training, provided availability and split boundaries are respected. RL rewards depend on actions and holdings, and TD propagation assigns delayed credit. Matching feature names does not match these signals.

Historical replay is partly interactive. A fixed market history can still support sequential action-dependent holdings and rewards. Retained Q-values and changing exploration can produce different trajectories on repeated episodes. Those passes do not create independent market histories, and immutable recorded prices do not model the agent's causal effect on future order flow.

Fair comparison requires aligned conditions. Align market information, timestamp availability, access to holdings, execution rules, capital, leverage, periods, costs, and metric conventions. Apply the same development discipline to manual rules and learned strategies. Report cumulative return, risk, drawdown, turnover, and benchmark-relative results on new data; one winning day, a small gross edge, or the strongest training score does not establish reliable economic superiority.

Every source group

G1 | Q-Learner vs. Decision-Tree Strategy Learner

ML2-D8G1Q1 p. 537 / ML2-D8G1Q2 p. 537 / ML2-D8G1Q3 p. 538

A tree fits supplied targets; a Q-learner updates values from rewards and transitions. Document outputs, exploration, training passes, seeds, and stopping rules. Treat identical market indicators as only one controlled aspect of the comparison.

Watch the wording: An RL action is not necessarily a numerical forecast. Repeated RL episodes do not guarantee a globally optimal trading policy.

G2 | Direct Policy Learning vs. Predict-Then-Trade

ML2-D8G2Q1 p. 539 / ML2-D8G2Q2 p. 539 / ML2-D8G2Q3 p. 540

Separate the predictor from the rule converting its output to a position. Permit costs and holdings to influence the execution rule when they matter. Evaluate net portfolio results instead of assuming forecast accuracy settles the comparison.

Watch the wording: A no-trade decision can be economically correct. Predict-then-trade systems can incorporate costs too.

G3 | Labels, Rewards, and Learning Signals

ML2-D8G3Q1 p. 540 / ML2-D8G3Q2 p. 541 / ML2-D8G3Q3 p. 542

Read label horizon, thresholds, and embedded cost assumptions. Read reward timing, dependence on positions, and discounting. Treat label construction and reward design as strategy choices.

Watch the wording: Future outcomes used only as training targets are not automatically leakage. A negative initial reward does not establish a negative cumulative action value.

G4 | Batch Learning vs. Interactive Learning

ML2-D8G4Q1 p. 542 / ML2-D8G4Q2 p. 543 / ML2-D8G4Q3 p. 544

Fixed prices can coexist with action-dependent portfolio transitions. Retaining learned values across resets can change later training actions. Describe which simulator variables respond to the agent.

Watch the wording: Repeated passes through one history are not independent market samples. A fixed impact formula does not reproduce every live market feedback effect.

G5 | Shared Features and Fair Inputs

ML2-D8G5Q1 p. 544 / ML2-D8G5Q2 p. 545 / ML2-D8G5Q3 p. 546

Align mathematical feature definitions, windows, and availability timestamps. Compare complete information access, including holdings used by downstream decision rules. Finalize manual thresholds using development data just as for learned parameters.

Watch the wording: Identical feature names do not imply identical information. A manual rule can overfit a test chart.

G6 | Fair Strategy Comparison

ML2-D8G6Q1 p. 546 / ML2-D8G6Q2 p. 547 / ML2-D8G6Q3 p. 547

Use aligned dates, starting capital, allowed leverage, execution, and costs. Calculate the same metric set under the same conventions for each strategy. Include a common benchmark and consider both risk and return.

Watch the wording: Dollar profit alone confounds capital and permitted exposure. Common dates do not make incompatible ranking metrics comparable.

G7 | Manual, Supervised, and RL Strategy Evaluation

ML2-D8G7Q1 p. 548 / ML2-D8G7Q2 p. 549 / ML2-D8G7Q3 p. 549

Evaluate the selected systems on unseen periods under matched conditions. Distinguish win frequency, statistical reliability, and economic effect size. Investigate sensitivity to regimes, randomness, turnover, and costs.

Watch the wording: One correct short trade cannot establish universal algorithm superiority. Statistical significance would not by itself make a tiny net gain worthwhile.

Formulas and conditions

Prediction-to-position rule

H_target = +L if forecast > τ_buy; -L if forecast < τ_sell; otherwise use the specified cash/hold rule

Threshold and cash-versus-hold conventions are separate strategy parameters. Holdings-dependent bands can incorporate turnover costs.

Net economic difference

Δ_net = (gross_A - costs_A) - (gross_B - costs_B)

Use matched capital, horizons, exposure constraints, and consistent cost accounting before comparing.

Window win rate

win rate = wins / evaluated windows

A win rate alone omits magnitude, uncertainty, dependence across windows, and transaction costs.

Work it through

The threshold changes trades, not the forecast

ML2-D8G2Q2 / pool p. 539

A fixed model forecasts a 0.15% next-day return. Its rule initially trades every positive prediction; a revised rule requires more than 0.20%.

The forecast remains 0.15%. The zero-threshold rule trades; the 0.20% rule does not. If the modeled full trade cost is 0.20%, the simple expected net gain is -0.05% before other effects.

Result: A cost-aware decision layer can improve alignment without refitting the predictor.

A small win count is not an economic edge

ML2-D8G7Q2 / pool p. 549

A learner wins in 11 of 20 windows. Its average gross advantage is 0.03 percentage points per window, but extra costs average 0.05 points.

Win rate = 11/20 = 55%. Average net difference = 0.03 - 0.05 = -0.02 percentage points per window. Win frequency does not supply a significance test or a positive net effect.

Result: The reported gross advantage disappears after costs.

ML2-D9 / MACHINE LEARNING / 21 QUESTIONS

Deep Learning and Neural-Network Foundations

Core understanding

A network is a parameterized computation. A dense unit combines inputs through weights and a bias, then applies an activation. Width counts units within a layer; depth counts successive layers. Equal total hidden-unit counts do not imply equal connectivity, parameter count, or represented function. An output's interpretation depends on its target and loss, not merely whether it is a scalar.

Nonlinearity makes depth expressive. Composing affine maps without nonlinear activations remains one affine map, regardless of depth. Nonlinear activations allow richer learned representations. Sigmoid units with extreme pre-activations can saturate and yield small local derivatives, weakening earlier gradients. A visually interpretable hidden pattern does not by itself establish a causal financial mechanism.

Forward prediction is distinct from learning. A forward pass applies learned transformations to inputs and produces a score or forecast. Ordinary inference needs neither target labels nor a backward pass. Fixed parameters and deterministic evaluation operations produce the same output for the same input. Consistent preprocessing and correct evaluation mode matter; a downstream trade threshold can change while the network score stays fixed.

Loss and optimization specify what improves. The loss compares predictions with targets; MSE penalizes large errors quadratically, while classification losses address class probabilities and labels. Backpropagation computes gradients by the chain rule, and an optimizer uses them to update weights. The learning rate controls step size. Lower forecast loss does not directly optimize Sharpe ratio, drawdown, costs, or portfolio decisions.

Training passes are not evidence of generalization. One epoch processes the training dataset once; mini-batches permit multiple parameter updates within an epoch. Batch size affects gradient noise, memory, and compute, rather than network width. Falling training loss alongside rising validation loss suggests overfitting. Dropout, weight penalties, early stopping, and simpler baselines can improve model selection, but none guarantees profitable future trades.

Every source group

G1 | Neural-Network Structure

ML2-D9G1Q1 p. 552 / ML2-D9G1Q2 p. 553 / ML2-D9G1Q3 p. 554

Distinguish input, hidden, and output layers and specify their connectivity. Count depth separately from layer width. Interpret outputs according to the modeled target and objective.

Watch the wording: Equal hidden-unit totals do not imply equal parameter counts. Adding units that contribute nothing need not change the function.

G2 | Activations and Nonlinearity

ML2-D9G2Q1 p. 554 / ML2-D9G2Q2 p. 555 / ML2-D9G2Q3 p. 556

An affine map is a weighted sum plus bias; stacked affine maps remain affine. Nonlinear activations allow nonlinear composition. Saturated activations can reduce gradient flow without stopping forward computation.

Watch the wording: Bias terms alone do not introduce nonlinear input-output behavior. Hidden-unit responses are not proof of causal economic meaning.

G3 | Forward Propagation and Prediction

ML2-D9G3Q1 p. 556 / ML2-D9G3Q2 p. 557 / ML2-D9G3Q3 p. 557

Apply fitted preprocessing and the learned parameters during the forward pass. Ordinary inference does not require labels, gradient calculation, or weight updates. Separate the network score from the rule converting it to a trade.

Watch the wording: A threshold change need not change the weights or score. A 0.70 score does not prove calibration in every market regime.

G4 | Loss Functions and Learning Objectives

ML2-D9G4Q1 p. 558 / ML2-D9G4Q2 p. 559 / ML2-D9G4Q3 p. 559

Match the loss to the output and target: continuous value or class probability. Assess how large errors and weighting influence the objective. Evaluate portfolio economics separately from statistical prediction loss.

Watch the wording: MSE does not directly maximize Sharpe ratio or minimize drawdown. Identical input features do not force identical learning objectives.

G5 | Gradient Descent and Backpropagation

ML2-D9G5Q1 p. 560 / ML2-D9G5Q2 p. 560 / ML2-D9G5Q3 p. 561

Compute gradients through the full computation graph using the chain rule. A gradient-descent optimizer moves opposite the gradient, scaled by a learning rate. Use training and validation evidence to assess unstable or slow optimization.

Watch the wording: Backpropagation calculates derivatives; the optimizer applies updates. One zero derivative is not proof of a global optimum.

G6 | Epochs, Batches, and Training Dynamics

ML2-D9G6Q1 p. 562 / ML2-D9G6Q2 p. 562 / ML2-D9G6Q3 p. 563

One epoch is a complete dataset pass under the stated sampling scheme. Mini-batch learning can update once per batch, many times per epoch. Track training and validation curves and select checkpoints by the development protocol.

Watch the wording: Batch size does not count neurons. More epochs can increase memorization and worsen held-out performance.

G7 | Capacity, Overfitting, and Regularization

ML2-D9G7Q1 p. 564 / ML2-D9G7Q2 p. 564 / ML2-D9G7Q3 p. 565

Assess capacity relative to data quantity, quality, and deployment shift. Use dropout, L2 penalties, or early stopping when supported by validation. Compare simpler baselines and account for computational requirements.

Watch the wording: Near-zero training loss does not guarantee durable market structure. No universal million-observation cutoff defines when neural networks can be used.

Formulas and conditions

Dense layer

$z^{(\ell)} = W^{(\ell)}h^{(\ell-1)} + b^{(\ell)}$; $h^{(\ell)} = \varphi(z^{(\ell)})$

Dimensions and activation choices define the architecture. The output layer may use a different activation or no nonlinear activation.

Dense parameter count

$\text{parameters} = n_{in} \times n_{out} + n_{out}$

Assumes every input connects to every output and there is one trainable bias per output unit.

MSE and L2-regularized objective

$MSE = \frac{1}{N}\sum_i (\hat{y}_i - y_i)^2$; $L_{total} = L_{data} + \lambda \sum_j w_j^2$

Penalty factors such as 1/2 and whether biases are penalized are implementation conventions. Choose λ using development data.

Gradient descent

$\theta_{new} = \theta_{old} - \eta \nabla_\theta L$

η is the optimizer learning rate; a large step need not reduce a nonlinear loss.

Batches per epoch

$\text{batches} = \text{ceil}(N/B)$

Assumes one pass without replacement, retaining the final partial batch. If partial batches are dropped, use floor(N/B).

Work it through

Count a small dense network

ML2-D9G1Q1 / pool p. 552

Three inputs feed four hidden units, which feed one scalar output. Include one bias per hidden and output unit.

Input-to-hidden weights: 3×4 = 12; hidden biases: 4. Hidden-to-output weights: 4×1 = 4; output bias: 1. Total = 12+4+4+1 = 21.

Result: Four hidden units form one hidden layer; the network has 21 parameters under these assumptions.

Compute an optimizer step

ML2-D9G5Q1 / pool p. 560

A scalar parameter is 2.0, its loss derivative is +3.0, and the gradient-descent learning rate is 0.1.

$\theta_{new} = 2.0 - 0.1 \times 3.0 = 1.7$. The positive derivative means increasing θ locally raises loss; gradient descent moves the other way.

Result: The new parameter is 1.7; this local step is not a global-optimality certificate.

Count updates, then choose the checkpoint

ML2-D9G6Q3 / pool p. 563

There are 10,000 observations, batches of 100, and one update per batch. Validation loss bottoms near epoch 40 but training continues to epoch 200.

Each epoch has 10,000/100 = 100 batches. By epoch 40 there are 4,000 updates; by epoch 200, 20,000. More updates do not override the validation evidence.

Result: A development-selected checkpoint near epoch 40 may generalize better than the final weights.

ML2-D10 / MACHINE LEARNING / 21 QUESTIONS

Sequential, Generative, and Responsible AI

Core understanding

Time order and availability define the forecasting problem. Lagged sequences can contain useful order information and are often statistically dependent, especially when windows overlap. Chronological evaluation better represents forecasting later periods from earlier evidence. Match feature timestamps to when data were released, including revisions; valid timing removes one failure mode but does not guarantee stability under a new regime.

Convolution and recurrence represent different structure. A temporal convolution reuses a local filter across ordered positions. Depth, kernel width, dilation, and pooling determine its effective receptive field; global pooling may remove precise position. A recurrent model carries hidden state forward and reuses update parameters through time. Long-range gradients can weaken, and recurrent state must be deliberately reset or initialized at independent sequence boundaries.

Attention connects positions through learned relevance. Self-attention can directly connect distant positions using data-dependent weights. A standard content-only attention mechanism needs positional information to explicitly represent sequence order; masks must enforce which information each forecast may access. Attention weights can reveal associations used in a computation but are not, by themselves, causal explanations or proof that irrelevant inputs are ignored.

Generative fluency is not factual or trading validation. LLMs can summarize filings, extract draft themes, and generate research text, but may omit qualifications or invent details. Retrieval adds candidate evidence and citations make it traceable; neither guarantees that the cited passage supports the claim. Verify consequential numbers, dates, and statements against sources, then separately test any trading decision rule after costs.

Responsible operation requires observable behavior and accountability. Aggregate accuracy can conceal subgroup errors and historical selection effects. Feature-attribution tools need fidelity and stability checks and do not establish causal effects. Deployment needs defined owners, version records, drift and incident monitoring, escalation, and intervention appropriate to the task. A successful backtest, a dollar limit, or unchanged code cannot guarantee that later decisions remain suitable.

Every source group

G1 | Time-Series and Sequential Modeling

ML2-D10G1Q1 p. 568 / ML2-D10G1Q2 p. 568 / ML2-D10G1Q3 p. 569

Preserve meaningful temporal order and use only information available at the forecast time. Recognize overlap and dependence between adjacent windows. Evaluate chronologically and assess performance under later regimes.

Watch the wording: A fixed-length window does not imply IID samples. A revised value’s reference date is not its publication date.

G2 | Convolutional Models for Financial Sequences

ML2-D10G2Q1 p. 570 / ML2-D10G2Q2 p. 570 / ML2-D10G2Q3 p. 571

Convolution shares local filter parameters across ordered positions and input channels. Stacking, dilation, and pooling determine access to wider context. Global pooling can retain motif strength while discarding exact location.

Watch the wording: Convolution is not invariant to arbitrary day reorderings. ML2-D10G2Q3 E has a reverse-key contradiction: more channels guarantee neither generalization nor causality; read the issue note.

G3 | Recurrent and Memory-Based Models

ML2-D10G3Q1 p. 572 / ML2-D10G3Q2 p. 572 / ML2-D10G3Q3 p. 573

A recurrent hidden state carries information from prior inputs. Reused recurrence parameters allow order-dependent sequence processing. Reset or deliberately initialize hidden state when starting an independent sequence.

Watch the wording: Longer input sequences do not guarantee learning of distant dependencies. Gated memory mechanisms help gradient flow but do not guarantee perfect memory.

G4 | Attention and Transformers

ML2-D10G4Q1 p. 574 / ML2-D10G4Q2 p. 574 / ML2-D10G4Q3 p. 575

Attention can form direct relationships between distant representations. Provide a valid representation of temporal position and enforce forecast-time information boundaries. Interpret attention as model association rather than proof of causation.

Watch the wording: Attention does not automatically exclude future targets. Removing explicit positions is not equivalent to adding recurrent memory.

G5 | Generative AI and Large Language Models

ML2-D10G5Q1 p. 575 / ML2-D10G5Q2 p. 576 / ML2-D10G5Q3 p. 577

Use generated text as an output to check, not an assumed verified source. Assess retrieval relevance and confirm whether citations entail the associated claims. Evaluate predictive and portfolio value

independently of formatting quality.

Watch the wording: A syntactically valid BUY label demonstrates no trading edge. Supplying a document does not guarantee every generated sentence is supported.

G6 | Hallucination, Verification, and Human Oversight

ML2-D10G6Q1 p. 577 / ML2-D10G6Q2 p. 578 / ML2-D10G6Q3 p. 579

Verify material claims and numbers against the cited original source. Use automated checks and human review with explicit ownership for consequential outputs. Provide anomaly escalation and intervention mechanisms for autonomous actions.

Watch the wording: Polished writing is not a reliable confidence measure. A position or dollar cap addresses only some failure modes.

G7 | Bias, Explainability, and Responsible Deployment

ML2-D10G7Q1 p. 579 / ML2-D10G7Q2 p. 580 / ML2-D10G7Q3 p. 580

Inspect relevant subgroup performance as well as aggregate metrics. Assess explanation stability and fidelity without assuming causality. Maintain versions, monitoring thresholds, owners, and incident-response procedures.

Watch the wording: Removing protected attributes does not remove correlated proxies. A one-time backtest does not establish perpetual suitability.

Formulas and conditions

One-dimensional convolution

$$y_t = \varphi(b + \Sigma_{j=0}^{k-1} \Sigma_{c=1}^{C} w_{j,c} x_{t+j,c})$$

Illustrates a width-k filter across C aligned channels. Select an input window and padding/mask so the computation cannot see unavailable future data.

Single-filter parameter count

$$\text{parameters} = kC + 1$$

One output filter with C input channels, width k, and one bias; the same parameters are reused at each position.

Recurrent update

$$h_t = \varphi(W_x x_t + W_h h_{t-1} + b)$$

A basic RNN illustration; gated models add state and gates. Reset boundaries should match independent sequences.

Scaled dot-product attention

$$\text{Attention}(Q,K,V) = \text{softmax}(QK^T / \sqrt{d_k} + M)V$$

M masks disallowed positions. Positional information and valid input construction remain necessary; attention weights are not causal effects.

Work it through

Convolution shares parameters

ML2-D10G2Q3 / pool p. 571

A single width-five filter spans three channels of a 20-day sequence. Use stride one, valid padding, and one bias.

Parameters = 5×3 + 1 = 16. Valid output positions = 20-5+1 = 16. The same 16 parameters are reused at all 16 positions. Global pooling could collapse those positions into one summary.

Result: Local parameter sharing is efficient; it does not guarantee generalization or preserve exact motif location.

Verify the actual debt reduction

ML2-D10G6Q2 / pool p. 578

A generated report says debt fell 40%. The cited filing instead reports a decline from $100 million to $86 million.

Debt change = $86M - $100M = -$14M. Reduction = $14M/$100M = 14%. A 40% reduction would imply $60M remaining, which the filing does not support.

Result: Correct the figure to 14% and verify the source before using the report.

Respect release time

ML2-D10G1Q2 / pool p. 568

Forecast day t+1 using information available at the close of t. An economic statistic describes month t but its revised value was released on t+3.

The statistic’s period label does not make the revision observable at t. Use the vintage actually released by t or omit the value. A causal attention mask cannot repair an input column already filled with the future revision.

Result: Enforce point-in-time availability before constructing model inputs.

QF2-D1 / QUANTITATIVE FINANCE / 21 QUESTIONS

Financial Time Series and Market Data Behavior

Core understanding

Use the right return representation. A price is a dollar level; a return measures change relative to invested capital. Equal dollar price changes can imply very different percentage gains. Simple returns compound by multiplying growth factors; log returns add across time. Normalizing prices to a common starting value makes cumulative price growth comparable but does not equalize risk, remove regime changes, or include omitted dividends.

Respect the information clock. A usable forecast must be computed from information available before the intended trade. Market closing times, publication times, lag direction, and the return measurement interval all matter. Trailing windows can respect this order; centered windows containing future observations cannot. Chronological or walk-forward evaluation more closely reproduces deployment than randomly mixing past and future observations.

Historical statistics can change. Short rolling windows react faster but are noisier; long windows are smoother but retain obsolete observations. Trends, structural breaks, changing competition, and shifting correlations can make old estimates unreliable. Returns may be more stable than price levels without being stationary. A precise estimate from a large old sample can still describe the wrong current regime.

Separate direction, magnitude, and realizability. Momentum is continuation; mean reversion is reversal toward a reference relationship. Both can occur at different horizons. Volatility clustering means large absolute returns cluster, even when return signs are hard to predict. Any signal must survive uncertainty, selection bias, its own decay, and implementation costs before it becomes an economically useful strategy.

Every source group

G1 | Prices, Returns, and Transformations

QF2-D1G1Q1 p. 588 / QF2-D1G1Q2 p. 588 / QF2-D1G1Q3 p. 589

Compare securities using percentage or total returns rather than dollar price changes when initial prices differ. Compound simple returns multiplicatively; add adjacent log returns only for positive prices. Normalized price P_t/P_0 measures relative price growth; use consistent dates and distribution adjustments.

Watch the wording: A high quoted share price does not establish high risk. Normalization changes scale, not future returns or stationarity.

G2 | Temporal Dependence and Lagged Relationships

QF2-D1G2Q1 p. 590 / QF2-D1G2Q2 p. 590 / QF2-D1G2Q3 p. 591

Autocorrelation measures association between a series and its lag; a negative lag-one estimate is consistent with average short-horizon reversal. Align actual information and trading times before interpreting a lead-lag relationship across markets. Use time-aware validation and investigate sampling uncertainty, bid-ask effects, and net trading costs.

Watch the wording: Correlation neither proves causation nor guarantees tomorrow's sign. More high-frequency rows or shuffled observations do not create independent evidence.

G3 | Rolling Windows and Time-Varying Statistics

QF2-D1G3Q1 p. 592 / QF2-D1G3Q2 p. 592 / QF2-D1G3Q3 p. 593

Short lookbacks favor responsiveness; long lookbacks favor smoothing and estimation stability. Rolling correlation changes the estimated diversification benefit and is not a permanent asset property. Use trailing data, a defined warm-up policy, and any signal lag needed to precede the traded return.

Watch the wording: Centered windows leak future information. Choosing the historically best window on the evaluation sample adds selection bias.

G4 | Trend, Momentum, and Mean Reversion

QF2-D1G4Q1 p. 594 / QF2-D1G4Q2 p. 594 / QF2-D1G4Q3 p. 595

Momentum and mean reversion are conditional hypotheses about a specified horizon, not universal laws. A pair trade may rely on mean reversion of a spread even when neither stock price is mean reverting. Check economic relationships, events, liquidity, and regime changes before treating an extreme move as continuation or reversal.

Watch the wording: A wide spread does not guarantee convergence after a structural break. A large price jump alone does not identify its economic mechanism.

G5 | Stationarity and Nonstationarity

QF2-D1G5Q1 p. 596 / QF2-D1G5Q2 p. 596 / QF2-D1G5Q3 p. 597

A changing mean, variance, or dependence structure can invalidate stationary historical estimates. Trending series can produce impressive in-sample relationships without useful predictive content. Compare rolling, subperiod, and forward results; adapting more quickly also increases estimation noise.

Watch the wording: More obsolete observations do not repair a structural break. Positive full-sample return does

not imply positive current expected return.

G6 | Volatility Clustering and Regime Behavior

QF2-D1G6Q1 p. 598 / QF2-D1G6Q2 p. 599 / QF2-D1G6Q3 p. 599

Persistence in absolute or squared returns can coexist with little autocorrelation in signed returns. Volatility targeting can increase exposure in calm markets and leave the portfolio overexposed when volatility jumps. Examine stressed and reversing regimes because correlation and volatility can rise together.

Watch the wording: Predictable volatility does not imply predictable return direction. Regimes defined after seeing losses can introduce selection bias.

G7 | Information Decay and Signal-to-Noise

QF2-D1G7Q1 p. 600 / QF2-D1G7Q2 p. 601 / QF2-D1G7Q3 p. 601

Match the forecast target, holding period, and actual execution time to the signal's decay horizon. Compare realizable net value, including latency and turnover, rather than only gross forecast magnitude. Testing many candidate indicators increases false discovery risk; repeated test-set tuning consumes the holdout.

Watch the wording: A significant 20-minute effect does not establish next-day profitability. No economic rationale raises scrutiny but does not mathematically prove zero predictive value.

Formulas and conditions

Simple and compounded returns

$r_t = P_t / P_{(t-1)} - 1$; $R_{0:T} = \prod(1 + r_t) - 1$

Prices must be consistently adjusted; price return omits distributions unless incorporated.

Log return

$\ell_t = \ln(P_t / P_{(t-1)})$; $\Sigma\ell_t = \ln(P_T / P_0)$; $R = \exp(\Sigma\ell_t) - 1$

Positive prices; logs are additive across adjacent time periods, not a change in economic wealth.

Normalized price

$N_t = P_t / P_0$; cumulative price return = $N_t - 1$

Common start date and consistent treatment of dividends, splits, and other adjustments.

Lag-k autocorrelation

$\rho_k = \mathrm{Corr}(r_t, r_{(t-k)})$

An estimated average relationship; sampling uncertainty and microstructure can matter.

Trailing sample statistics

$\mu_t = (1/n)\ \Sigma\ r_{(t-j)}$; $s_t^2 = [\Sigma(r_{(t-j)} - \mu_t)^2] / (n-1)$, $j = 0,\ldots,n-1$

A full trailing sample with $n > 1$; information through t must be available before the trade.

Work it through

Equal dollar moves, unequal returns

QF2-D1G1Q1 / pool p. 588

A $25 stock and a $500 stock each rise $5. Which performs better for equal-dollar positions?

The returns are 5/25 = 20% and 5/500 = 1%. An equal $1,000 investment earns $200 versus $10 before costs.

Result: The lower-priced stock earns the higher percentage return; share price itself does not measure risk.

Compounding does not cancel

QF2-D1G1Q2 / pool p. 588

A $100 investment earns +10% and then -10%.

Ending wealth = 100 × 1.10 × 0.90 = $99. Simple-return sum = 0%; cumulative return = -1%. Log-return sum = ln(0.99) ≈ -0.01005.

Result: The actual loss is $1 regardless of return representation.

A valid statistic at an unusable time

QF2-D1G7Q1 / pool p. 600

A news effect is strongest within 20 minutes, but execution occurs at the next day's close.

The relevant question is how much predictive return remains after the execution time. The original 20-minute event study does not measure that remaining return; test the tradable horizon with costs.

Result: Statistical evidence at the original horizon is insufficient for this implementation.

QF2-D2 / QUANTITATIVE FINANCE / 21 QUESTIONS

Technical Indicators and Signal Construction

Core understanding

An indicator summarizes the past. Moving averages smooth price history; momentum compares endpoints; bands locate price relative to a rolling center and dispersion; oscillators summarize recent strength on a bounded scale. Each construction describes observations. A complete strategy additionally needs a hypothesis, entry and exit rules, position sizes, risk limits, and feasible execution.

Responsiveness always has a cost. Short windows react earlier but respond more strongly to noise and can cause whipsaws. Long windows smooth more but can remain anchored to a regime that has already ended. Exponential weighting emphasizes recent observations without predicting a move before it occurs. Window selection should be tested for stability rather than justified by one sharp in-sample optimum.

Thresholds are choices, not guarantees. A price above an upper Bollinger Band may support a breakout hypothesis or a mean-reversion hypothesis. An overbought oscillator can stay overbought throughout an uptrend. Requiring confirmation may reduce some premature entries but delays action and does not eliminate false signals. Interpret thresholds in the asset, horizon, and regime where they will be used.

Combine information deliberately. Percentage returns and z-scores help compare different scales; ranks retain order while discarding magnitude differences. None of these transformations creates predictive information or stationarity. Multiple price-trend indicators may largely repeat the same information. Conflicting signals require a predefined combination or regime rule whose net performance is evaluated out of sample.

Every source group

G1 | Moving Averages and Trend Measures

QF2-D2G1Q1 p. 605 / QF2-D2G1Q2 p. 605 / QF2-D2G1Q3 p. 606

An n-period SMA gives equal weight to observations inside its current window; an EMA conventionally gives more weight to recent observations. A short average typically reacts faster and whipsaws more; a long average typically smooths more and lags more. A crossover compares fast and slow averages and still requires explicit sizing, exits, costs, and execution rules.

Watch the wording: EMA does not guarantee advance knowledge or greater profitability. A fast-over-slow crossover does not require current price to be above both averages.

G2 | Momentum and Rate-of-Change Indicators

QF2-D2G2Q1 p. 607 / QF2-D2G2Q2 p. 608 / QF2-D2G2Q3 p. 608

Momentum P_t/P_(t-L) - 1 measures the endpoint return over lookback L. The same endpoint return can conceal different paths, volatility, and drawdowns. A large old move leaving the window can sharply change the indicator even when current price barely changes.

Watch the wording: Positive trailing momentum does not guarantee positive future return. A reversal can begin while a long-horizon momentum rank is still high.

G3 | Bollinger Bands and Volatility-Based Indicators

QF2-D2G3Q1 p. 609 / QF2-D2G3Q2 p. 610 / QF2-D2G3Q3 p. 610

Bands are a rolling center plus or minus a chosen multiple of rolling standard deviation. With the multiplier fixed, widening bands indicate higher measured dispersion, not automatically higher expected return. A lower-band buy embeds a mean-reversion assumption; persistent trends can keep price near or beyond a band.

Watch the wording: A two-standard-deviation band is not an impossible-to-breach boundary. An upper-band breach alone does not establish that shorting is correct.

G4 | Relative Strength and Oscillator-Style Indicators

QF2-D2G4Q1 p. 611 / QF2-D2G4Q2 p. 612 / QF2-D2G4Q3 p. 613

Overbought and oversold labels identify regions of a chosen oscillator, not certain reversal points. A return below an upper threshold seeks confirmation of weakening strength and can delay entry. Divergence compares price and oscillator behavior; its interpretation depends on construction and lookback.

Watch the wording: A bounded indicator does not bound the underlying asset's losses. Neither divergence nor confirmation replaces a complete tested trading rule.

G5 | Indicator Normalization and Comparability

QF2-D2G5Q1 p. 613 / QF2-D2G5Q2 p. 614 / QF2-D2G5Q3 p. 615

Divide dollar changes by starting prices for comparable percentage moves. A z-score measures distance from the selected mean in units of the selected standard deviation. Ranks preserve ordering and reduce extreme magnitude dominance but lose numerical spacing.

Watch the wording: Standardization does not eliminate nonstationarity or make a forecast valid. Top-ranked assets need not have positive absolute expected returns.

G6 | Lookback Windows, Lag, and Responsiveness

QF2-D2G6Q1 p. 615 / QF2-D2G6Q2 p. 616 / QF2-D2G6Q3 p. 616

The effective memory of an indicator determines how quickly old-regime data leave it. Faster reaction may raise noise sensitivity and turnover; smoothing may delay recognition of a real change. Assess nearby lookbacks and untouched periods to distinguish a stable region from a lucky parameter peak.

Watch the wording: Slow adjustment is not automatically a software bug. The exact best in-sample lookback is not automatically the best live choice.

G7 | Combining Indicators into Trading Signals

QF2-D2G7Q1 p. 617 / QF2-D2G7Q2 p. 618 / QF2-D2G7Q3 p. 619

Inspect correlation and incremental predictive value before treating multiple indicators as independent confirmations. Specify how conflicting signals map to one executable position or to a decision to wait. Validate the combined signal, thresholds, risk controls, and net returns as a whole. Align component scales and directions before averaging; a composite crossing zero repeatedly can generate costly churn.

Watch the wording: Three trend indicators do not necessarily provide three independent sources of information. Agreement or disagreement alone does not prove a profitable strategy.

Formulas and conditions

Simple moving average

$SMA_t(n) = (1/n) \Sigma P_{(t-j)}, j = 0,\ldots,n-1$

A trailing window; define warm-up behavior and the decision timestamp.

Exponential moving average

$EMA_t = \alpha P_t + (1-\alpha)EMA_{(t-1)}$; commonly $\alpha = 2/(n+1)$

Specify initialization and convention; nominal n does not make EMA identical to an n-period SMA.

Momentum / rate of change

$M_t(L) = P_t / P_{(t-L)} - 1$

Positive comparable prices; an endpoint statistic, not a forecast guarantee.

Bollinger Bands

Upper $= \mu_t + k\sigma_t$; Lower $= \mu_t - k\sigma_t$; width $= 2k\sigma_t$

μ_t and σ_t use the same defined trailing price window; k is chosen, often 2.

Normalized location

$z = (x - \mu)/\sigma$; $\%B = (P - Lower)/(Upper - Lower)$

σ and band width must be nonzero; z uses standard deviations, whereas %B is band position and may exceed [0,1].

Lecture Bollinger value

$BB = (P - SMA)/(2\sigma) = z/2$; for $k = 2$, $\%B = (BB + 1)/2$

The course uses BB = -1 at the lower two-sigma band, 0 at the mean, and +1 at the upper band. This differs from z-score and percent-B; all require a nonzero denominator.

Work it through

Bollinger arithmetic before interpretation

QF2-D2G3Q1 / pool p. 609

Rolling mean is \$100, standard deviation is \$4, k = 2, and price closes at \$109.

Upper = 100 + 2×4 = \$108; lower = \$92. Price is \$1 above the upper band. The z-score is (109-100)/4 = 2.25, lecture BB = 9/8 = 1.125, and %B = 17/16 = 1.0625.

Result: The arithmetic establishes a breach; continuation versus reversal still needs a tested hypothesis.

Compare unusualness, not raw size

QF2-D2G5Q2 / pool p. 614

Market A has signal 4, mean 1, standard deviation 1.5; B has signal 8, mean 6, standard deviation 0.5.

$z_A = (4-1)/1.5 = 2$. $z_B = (8-6)/0.5 = 4$. B is farther above its own reference mean in standardized units.

Result: B is more unusual relative to its estimated distribution; this is not proof of a larger future profit.

Percentage momentum changes the ranking

QF2-D2G5Q1 / pool p. 613

A \$20 stock gains \$2, while a \$400 stock gains \$8.

The raw change favors the second stock 8 to 2, but returns are 2/20 = 10% and 8/400 = 2%.

Result: The first stock ranks higher by percentage momentum.

QF2-D3 / QUANTITATIVE FINANCE / 21 QUESTIONS

Alpha Research and Formulaic Signals

Core understanding

Define exactly what the alpha predicts. An alpha can be an expected benchmark-relative return, a relative ranking, or another score. State its units, benchmark, horizon, and information time before interpreting its magnitude. A high score is a forecast rather than a realized return. Forecast confidence can differ even when point estimates match, and unitless scores need calibration before proportional sizing.

Absolute and relative forecasts differ. A time-series signal evaluates an asset against its own history or predicts its own future direction. A cross-sectional signal orders assets at the same date. The weakest-ranked asset can still have positive expected absolute return, and a directional rule can make every position long. Market neutrality comes from portfolio construction and exposure control, not from the mere use of ranks.

Transformations need an economic purpose. Ranks reduce scale dominance but discard distances; z-scores standardize relative to estimated location and dispersion; winsorization caps extremes; neutralization removes fitted exposure to a chosen variable. These operations do not create skill. Formulaic combinations must specify units, signs, missing values, near-zero denominators, and lookbacks, then justify added complexity through independent evidence.

Measure what remains after implementation. Predictive strength decays with signal age, while refreshing positions incurs turnover and costs. Correlated alphas add less information than their count suggests, though averaging nearby specifications may improve robustness. Crowding can accelerate price adjustment and synchronize exits. Evaluate a composite's marginal contribution, stability, execution horizon, and capacity rather than extrapolating gross backtest alpha.

Every source group

G1 | Defining and Interpreting Alpha Signals

QF2-D3G1Q1 p. 622 / QF2-D3G1Q2 p. 622 / QF2-D3G1Q3 p. 623

Specify whether a score is an expected active return, a rank, or an uncalibrated number, and state its horizon. Judge a forecast using subsequent outcomes over the stated horizon, not one unrelated prior-day loss. Sizing can reflect forecast confidence, risk, and constraints in addition to the point estimate.

Watch the wording: A rank of 80 does not imply twice the expected return of rank 40. A confident forecast does not guarantee a positive realized return.

G2 | Time-Series vs. Cross-Sectional Signals

QF2-D3G2Q1 p. 624 / QF2-D3G2Q2 p. 624 / QF2-D3G2Q3 p. 625

Time-series direction and cross-sectional relative attractiveness answer different questions. A bottom-ranked asset can be shorted in a relative strategy even when its absolute forecast is positive. Align scales and horizons when combining signal types, then explicitly enforce portfolio exposures.

Watch the wording: Cross-sectional ranks do not inherently guarantee dollar or market neutrality. Using one signal of each type does not guarantee independence.

G3 | Ranking, Scaling, and Signal Transformation

QF2-D3G3Q1 p. 626 / QF2-D3G3Q2 p. 626 / QF2-D3G3Q3 p. 627

Percentile ranks preserve within-group ordering; within-industry ranking can reduce unintended industry dominance. Z-scores are sensitive to estimated means, dispersion, and outliers; winsorization caps chosen extremes. Residualizing against size removes the fitted linear size-related component, which may include useful information.

Watch the wording: Ranks do not preserve expected-return distances. Winsorization and neutralization do not guarantee clean data or improved out-of-sample alpha.

G4 | Formulaic Alpha Construction

QF2-D3G4Q1 p. 628 / QF2-D3G4Q2 p. 628 / QF2-D3G4Q3 p. 629

Fully define formula units, sign behavior, lookbacks, missing values, and denominator safeguards. Dividing by volatility changes scaling and can produce unstable values when volatility approaches zero. Require independent incremental benefit from added transformations and test nearby parameter combinations.

Watch the wording: Multiplying plausible indicators does not establish predictiveness. Many nonlinear variants increase the chance of selecting noise; monotonic variants may preserve the same ranking.

G5 | Signal Decay and Holding Period

QF2-D3G5Q1 p. 630 / QF2-D3G5Q2 p. 631 / QF2-D3G5Q3 p. 631

Estimate predictive value by signal age and align entry, exit, and return targets with that decay. A longer hold can reduce costs while leaving capital exposed after demonstrated alpha has disappeared. Choose rebalancing using net benefit, liquidity, and decay; evaluate staggered updates when appropriate.

Watch the wording: Strong 30-minute IC does not establish a once-daily strategy. The horizon with the highest gross IC need not maximize net returns.

G6 | Combining Multiple Alpha Signals

QF2-D3G6Q1 p. 632 / QF2-D3G6Q2 p. 633 / QF2-D3G6Q3 p. 633

Combine signals using both predictive reliability and incremental information beyond existing signals. Correlation, estimation uncertainty, turnover, and costs can change sensible weights from a simple IC-proportional rule. Averaging overlapping lookbacks can improve parameter robustness without adding equivalent independent breadth.

Watch the wording: Equal signal weights do not imply equal portfolio risk contributions. The weakest standalone alpha may still help if it is complementary and inexpensive.

G7 | Crowding, Replication, and Realizable Alpha

QF2-D3G7Q1 p. 634 / QF2-D3G7Q2 p. 635 / QF2-D3G7Q3 p. 635

Replication can bring price adjustment forward and reduce the remaining return available to later traders. Larger capital and crowded positions can increase impact, capacity pressure, and synchronized exit risk. Use economic rationale, point-in-time testing, and cross-period robustness to guide further research.

Watch the wording: Historical gross alpha is not guaranteed to persist after widespread adoption. A lack of rationale is not mathematical proof of no signal, but a spectacular searched backtest needs stronger validation.

Formulas and conditions

Active return target

r_active = r_asset - r_benchmark

Specify benchmark and forecast horizon; a score may only rank this quantity rather than estimate it in return units.

Cross-sectional z-score

z_i,t = (x_i,t - mean_t(x)) / sd_t(x)

Uses the stated same-date universe; extreme data can distort both location and dispersion.

Linear neutralization

x_i = a + b·size_i + ε_i; neutralized signal = ε_i

Residual orthogonality is an in-sample fitted linear property, not guaranteed future neutrality.

Illustrative price-volume alpha

α_t = M_t(20) × [V_t / mean_t(V)] / σ_t

Define all windows, positive denominator floors, units, and timestamps; this formula is a hypothesis, not established alpha.

Composite alpha

α_composite = Σ w_j z_j

Standardization aligns scale; weights still require attention to reliability, dependence, decay, and implementation cost.

Work it through

Read a formula’s sensitivity

QF2-D3G4Q1 / pool p. 628

A proposed alpha multiplies 20-day return by relative volume and divides by volatility. Illustrate with return 0.10, relative volume 1.5, and volatility 0.20.

Score = 0.10×1.5/0.20 = 0.75. Holding other values fixed, volatility 0.10 gives 1.50. If the denominator approaches zero, the raw score becomes unstable.

Result: The calculation explains scale and the need for a denominator safeguard; 0.75 is not a 75% expected return.

Relative shorts during a broad rally

QF2-D3G2Q2 / pool p. 624

All 100 stocks have positive forecasts. A relative strategy buys the top decile and shorts the bottom decile.

In an illustration, top-decile stocks gain 8% and bottom-decile stocks gain 3%. A $100 long and $100 short produce $8-$3 = $5 before costs, although both groups rose.

Result: Relative spread performance can be positive while the short leg loses; dollar neutrality does not guarantee factor neutrality.

Weakest standalone can still diversify

QF2-D3G6Q2 / pool p. 633

Three standardized alphas have IC estimates 0.06, 0.04, and 0.02; the first two are highly correlated and the third trades less.

IC-proportional weights normalize to 50%, 33.3%, and 16.7%, but those weights ignore redundancy, estimation uncertainty, and costs. Test the third signal’s incremental net contribution rather than discarding it.

Result: The largest IC is not sufficient to determine the best composite weight.

QF2-D4 / QUANTITATIVE FINANCE / 21 QUESTIONS

Fundamental Law of Active Portfolio Management

Core understanding

IC measures forecast association. The information coefficient is the correlation between forecasts and subsequent outcomes defined on a consistent universe and horizon. A small positive IC can be useful if persistent and applied across sufficiently independent opportunities. It is not a hit rate or a return. A negative IC can contain inverted information, while zero linear correlation does not rule out every nonlinear relationship.

IR measures active efficiency. The information ratio divides average benchmark-relative return by the volatility of that active return, called tracking error. Sharpe instead compares total portfolio return with a risk-free reference and uses total volatility. Equal historical IR values do not establish equal reliability: sample length, benchmark choice, concentration, costs, and regimes affect interpretation.

Independent breadth is the multiplier. The simplified Fundamental Law relates IR to IC times the square root of breadth. Skill enters linearly; independent opportunity count enters through a square root. Breadth can come from assets or time, but correlated securities, overlapping horizons, and near-duplicate signals reduce effective breadth. Counting trades or formula columns mechanically overstates the independent decisions.

A design relationship, with assumptions. The simplified law assumes sufficiently stable skill, comparable independent opportunities, and efficient translation of forecasts into positions. More coverage may dilute expertise; automation may replicate the same bet. Constraints, trading costs, impact, funding, and crowding can reduce realized performance. Test skill and dependence at the scale and horizon actually contemplated.

Every source group

G1 | Information Coefficient

QF2-D4G1Q1 p. 639 / QF2-D4G1Q2 p. 639 / QF2-D4G1Q3 p. 640

IC is correlation between matched forecasts and subsequent outcomes, not the percentage of correct trades. Judge IC together with effective sample size, regime coverage, consistent definitions, and uncertainty. Negating forecasts reverses their correlation with the same outcomes; investigate timing and labels before deploying a reversed signal.

Watch the wording: A larger point estimate from a tiny sample is not automatically more credible. IC near zero does not prove that all nonlinear information is absent.

G2 | Information Ratio

QF2-D4G2Q1 p. 641 / QF2-D4G2Q2 p. 642 / QF2-D4G2Q3 p. 642

IR uses mean active return divided by tracking error on a consistent frequency basis. Sharpe uses excess total return over the risk-free rate divided by total volatility. Consider benchmark suitability, sample duration, factor concentration, and net costs when comparing historical ratios.

Watch the wording: Higher active return alone does not imply a higher IR. Equal historical IR does not guarantee equal future skill or dollar profit.

G3 | Breadth and Number of Opportunities

QF2-D4G3Q1 p. 643 / QF2-D4G3Q2 p. 644 / QF2-D4G3Q3 p. 644

Breadth counts sufficiently independent opportunities to apply skill across assets and time. Correlated daily decisions can add much less effective breadth than the increase in raw forecast count. Automation expands processing capacity, but new decisions still need skill and realizability.

Watch the wording: Securities held, orders submitted, and independent bets are different quantities. Adding opportunities with no skill does not mechanically improve IR.

G4 | The Fundamental Law Relationship

QF2-D4G4Q1 p. 645 / QF2-D4G4Q2 p. 646 / QF2-D4G4Q3 p. 646

Use $IR \approx IC\sqrt{BR}$ only under the stated simplifying assumptions and consistent annual breadth. Doubling IC doubles predicted IR; doubling breadth multiplies it by $\sqrt{2}$. To double IR through breadth alone, quadruple effective breadth while preserving IC.

Watch the wording: A stronger IC can be outweighed by much lower breadth. The equation does not guarantee that new independent opportunities can be created costlessly.

G5 | Effective Breadth and Dependence

QF2-D4G5Q1 p. 647 / QF2-D4G5Q2 p. 648 / QF2-D4G5Q3 p. 648

Common factors can make many stock positions versions of one economic bet. Near-duplicate signals may change robustness or scaling without adding much independent information. Inspect stressed correlations and conditional exposures because effective diversification can decline in crises.

Watch the wording: Different company or asset-class names do not establish independence. Zero full-sample correlation does not imply independence in every regime.

G6 | Skill vs. Scale Tradeoffs

QF2-D4G6Q1 p. 649 / QF2-D4G6Q2 p. 650 / QF2-D4G6Q3 p. 650

Compare changes in both IC and effective breadth when expanding a strategy. A fourfold breadth increase offsets a halving of IC in the simplified relationship. Validate new universes and horizons because liquidity, data quality, and mechanisms can differ.

Watch the wording: Higher skill alone does not ensure higher predicted IR. Automation does not preserve the original IC by assumption.

G7 | Limitations and Practical Violations

QF2-D4G7Q1 p. 651 / QF2-D4G7Q2 p. 652 / QF2-D4G7Q3 p. 653

Re-estimate rolling and out-of-sample skill when competition or regimes change. Account for dependence, costs, constraints, capacity, and turnover before treating gross potential as attainable. Use gaps between predicted and live results to diagnose translation and implementation assumptions.

Watch the wording: A live IR below the simplified-law prediction need not mean the arithmetic was wrong. Unlimited liquidity and raw decision-count breadth are optimistic assumptions for crowded small-cap trading.

Formulas and conditions

Information coefficient

IC = Corr(forecast_i, subsequent active return_i)

Use matched horizon, universe, and information time; the effective sample may be smaller than the row count.

Information ratio

IR = mean(r_p - r_b) / sd(r_p - r_b)

Numerator and tracking-error denominator must use consistent horizons and annualization.

Simplified Fundamental Law

IR ≈ IC × √BR

BR is effective independent opportunities per year; assumes stable comparable skill and efficient implementation before practical frictions.

Comparative Fundamental Law

IR_new / IR_old = (IC_new / IC_old) × √(BR_new / BR_old)

Same assumptions and comparable definitions; nonzero old IC and IR.

Breadth required for a target

BR_required = (IR_target / IC)²

An algebraic implication for positive fixed IC, not proof the required breadth exists.

Work it through

Skill and breadth together

QF2-D4G4Q2 / pool p. 646

Strategy A has IC 0.04 and breadth 400. B has IC 0.08 and breadth 25.

IR_A = 0.04×20 = 0.80. IR_B = 0.08×5 = 0.40. B has twice the skill but one-sixteenth the breadth.

Result: A has twice the simplified predicted IR; actual net performance remains conditional on implementation.

Double IR through independent breadth

QF2-D4G4Q3 / pool p. 646

Current effective breadth is 64. IC remains unchanged and management wants twice the predicted IR.

The current multiplier is √64 = 8. Doubling it requires 16, so breadth must be $16^2 = 256$. Breadth 128 only gives a √2 ≈ 1.414 increase.

Result: Quadruple breadth from 64 to 256.

Return per active risk

QF2-D4G2Q1 / pool p. 641

Fund 1 earns 4% active return with 8% tracking error; fund 2 earns 3% with 3% tracking error.

IR_1 = 4/8 = 0.50. IR_2 = 3/3 = 1.00. The second has lower raw active return but twice the return per unit of active risk.

Result: Fund 2 has the higher information ratio under the stated consistent measurements.

QF2-D5 / QUANTITATIVE FINANCE / 21 QUESTIONS

Advanced Portfolio Construction

Core understanding

Forecasts need a position rule. A positive forecast can map to a long position, a negative forecast to a short, or an active view to an overweight or underweight. A linear rule preserves magnitude; a sign rule discards it. The same forecast produces different portfolios under different constraints. Risk, confidence, existing holdings, and feasibility determine how much of the forecast can sensibly be expressed.

Measure active exposure and total risk. Active weight equals portfolio weight minus benchmark weight. A negative active weight can arise from a positive holding below its benchmark allocation. Gross exposure sums absolute positions; net exposure sums signed positions. Neither zero net dollars nor equal capital weights establishes low or equal risk. Portfolio risk depends on covariances and common factor exposures.

Treat estimated optima with care. Expected returns and covariances are estimates. Small input changes that create extreme weight changes reveal sensitivity, not certainty. Shrinkage, regularization, caps, and scenario checks can reduce overreaction to noise, sometimes sacrificing an optimistic in-sample objective for more stable allocations. Robust allocation does not require equal weights in every asset.

The best feasible update may be no trade. Leverage, sector, short-sale, liquidity, position, and turnover limits define the feasible set. Rebalancing incurs costs on the difference between current and target holdings. When a small expected improvement costs more to implement than it is worth, a no-trade band or turnover penalty can favor retaining current positions. Turnover reduction remains subordinate to net value and risk needs.

Every source group

G1 | Translating Forecasts into Positions

QF2-D5G1Q1 p. 656 / QF2-D5G1Q2 p. 656 / QF2-D5G1Q3 p. 657

A linear forecast-to-position rule preserves relative magnitudes before constraints and normalization. A sign-based rule keeps direction but assigns equal raw magnitude within each sign. Long-only funds can express negative relative views through underweights while maintaining nonnegative holdings.

Watch the wording: Zero forecast does not imply the largest position. Equal gross exposure does not guarantee equal portfolio risk.

G2 | Position Sizing and Conviction

QF2-D5G2Q1 p. 658 / QF2-D5G2Q2 p. 658 / QF2-D5G2Q3 p. 659

Size using both forecast strength and risk; the stated independent-asset rule μ/σ^2 uses variance, not volatility. Confidence scaling can reduce a less reliable forecast's position without reversing its sign. Position caps can limit concentration driven by noisy point estimates and make an unconstrained solution infeasible.

Watch the wording: An 80% confidence label is not automatically an 80% probability of profit. Equal expected returns need not imply equal position sizes.

G3 | Active vs. Benchmark-Relative Weights

QF2-D5G3Q1 p. 660 / QF2-D5G3Q2 p. 660 / QF2-D5G3Q3 p. 661

Active weights subtract benchmark weights from actual holdings. A zero holding in a benchmark constituent is a negative active weight, not a short holding. Assess tracking error using active weights and covariance; active share measures composition difference.

Watch the wording: Zero net dollar exposure can still conceal high gross exposure and systematic risk. High active share does not guarantee positive active return.

G4 | Risk-Aware Portfolio Construction

QF2-D5G4Q1 p. 662 / QF2-D5G4Q2 p. 663 / QF2-D5G4Q3 p. 663

Portfolio variance depends on covariance among positions as well as each position's volatility. Equal dollar allocations or equal standalone volatilities do not imply equal contributions to portfolio risk. Inspect correlations and common exposures across strategy sleeves even when they hold different securities.

Watch the wording: Portfolio volatility is not generally the weighted average of asset volatilities. Negative estimated correlation does not guarantee perfect future offsetting returns.

G5 | Constraints and Implementability

QF2-D5G5Q1 p. 664 / QF2-D5G5Q2 p. 665 / QF2-D5G5Q3 p. 665

Apply gross, net, position, sector, short-sale, liquidity, and turnover constraints separately. A sector or liquidity limit can constrain a strong forecast even when individual signals were estimated independently. Daily participation limits apply to total trading; splitting a parent order into child orders does not reduce aggregate participation.

Watch the wording: A 100% net portfolio can violate a 150% gross-exposure cap. Removing a real constraint merely because it lowers backtest return does not make a portfolio implementable.

G6 | Forecast Uncertainty and Robustness

QF2-D5G6Q1 p. 666 / QF2-D5G6Q2 p. 667 / QF2-D5G6Q3 p. 667

Large weight changes after small forecast revisions warrant sensitivity analysis. Shrinkage and penalties can reduce concentration and turnover caused by uncertain inputs. Compare robust and unconstrained methods on rolling out-of-sample periods and plausible perturbations.

Watch the wording: An extreme optimizer weight is not evidence that the forecast is correct. Leverage amplifies the consequences of forecast error rather than removing uncertainty.

G7 | Optimization with Costs and Turnover

QF2-D5G7Q1 p. 668 / QF2-D5G7Q2 p. 669 / QF2-D5G7Q3 p. 669

Costs depend on trades from current holdings, so the starting portfolio is part of the optimization problem. A no-trade region can retain a position when incremental expected benefit is smaller than implementation cost. Entry or reversal thresholds and turnover penalties can reduce churn from weak noisy forecasts.

Watch the wording: Any deviation from a frictionless target does not automatically justify trading. Minimizing turnover at all costs can sacrifice valuable trades or needed risk reduction.

Formulas and conditions

Active weight and active share

$a_i = w_i - b_i$; Active Share = $\frac{1}{2} \Sigma|a_i|$

Active Share convention applies to comparable fully invested portfolio and benchmark weights; it is not tracking error.

Gross and net exposure

Gross = $\Sigma|w_i|$; Net = Σw_i

Weights are signed exposures relative to the same capital base; positive longs and negative shorts.

Portfolio and active risk

$\sigma_p^2 = w^T \Sigma w$; $TE^2 = (w-b)^T \Sigma (w-b)$

Σ is the covariance matrix on the chosen horizon; both estimates can change across regimes.

Two-asset portfolio variance

$\sigma_p^2 = w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1w_2\rho_{12}\sigma_1\sigma_2$

Use consistent horizons and correlations; diversification comes through the covariance term.

Illustrative independent-asset sizing

raw $w_i \propto \mu_i / \sigma_i^2$; confidence-scaled target $\propto c_i \times$ raw w_i

The question's simplified rule assumes independent assets and comparable forecasts; constraints and normalization come afterward.

Cost-aware allocation objective

$\text{maximize}_w [\mu^T w - (\lambda/2) w^T \Sigma w - C(w-w_{current})]$

Illustrative mean-variance objective subject to mandate constraints; C includes implementation costs and/or turnover penalties.

Work it through

Variance-based sizing

QF2-D5G2Q1 / pool p. 658

Two independent stocks each have 6% expected excess return; H has volatility 30% and L 15%.

Raw H score = $0.06/0.30^2 = 0.667$; L = $0.06/0.15^2 = 2.667$. L/H = 4. Halving volatility quarters variance.

Result: The stated μ/σ^2 rule assigns L four times H's raw size, before constraints.

Underweight is not short

QF2-D5G3Q1 / pool p. 660

Benchmark weights are A 5%, B 3%, C 0%; portfolio weights are 8%, 1%, 2%.

Active weights are +3, -2, and +2 percentage points. B is still a positive 1% holding despite its -2-point active weight. The remaining portfolio is not shown.

Result: B expresses a negative relative view without a short sale; full-portfolio active share cannot be computed from only these rows.

Correlation changes portfolio risk

QF2-D5G4Q1 / pool p. 662

Both assets have 20% volatility and equal 50% weights. Compare correlation +0.90 with -0.20.

Variance = $0.5 \times 0.20^2 \times (1+\rho)$. For $\rho = 0.90$, $\sigma \approx \sqrt{0.038} = 19.49\%$. For $\rho = -0.20$, $\sigma \approx \sqrt{0.016} = 12.65\%$.

Result: The lower-correlation pair has materially lower estimated risk despite identical individual volatilities.

QF2-D6 / QUANTITATIVE FINANCE / 21 QUESTIONS

Strategy Evaluation and Performance Attribution

Core understanding

Benchmark the economic claim. A useful benchmark reflects the strategy's opportunity set, market exposure, permissions, dates, and implementation assumptions. Cash, buy-and-hold, a market index, a simple manual rule, and factor benchmarks answer different questions. A long-only stock strategy beating cash may largely be earning market exposure; a near-market-neutral strategy needs more context than a comparison with a stock index.

Preserve independent evidence. Any data used to choose features, parameters, or trading rules belong to development. Changing a strategy after inspecting a holdout makes that period part of selection for the revision. Walk-forward evaluation can test a scheduled retraining process, provided every feature and fit respects information availability. Retraining does not make the resulting test returns statistically independent.

Describe both return and its path. Cumulative return records ending wealth; volatility measures periodic dispersion; drawdown records the loss from a previous peak. Sharpe, Sortino, and information ratios use different risk denominators and reference returns. Identical headline returns or ratios can conceal different paths, tails, sample lengths, and exposures. No single metric proves a strategy is suitable or robust.

Separate sources, costs, and confidence. Attribution distinguishes beta, sectors, factors, selection, and timing from unexplained performance. Net return subtracts the costs required by trading behavior. A strategy can have a positive mean without strong statistical evidence or adequate dollar profit. Test subperiods, regimes, parameter neighborhoods, cost assumptions, and dependence on a few extreme trades before treating a backtest as repeatable.

Every source group

G1 | Benchmark Selection

QF2-D6G1Q1 p. 673 / QF2-D6G1Q2 p. 673 / QF2-D6G1Q3 p. 674

Choose benchmarks that match the strategy's market exposure, opportunity set, and permissions. Use identical dates, initial capital, and consistent cost assumptions to compare learned, manual, and passive rules. Use multiple relevant references when they isolate different claims about value added.

Watch the wording: Beating cash by 9 percentage points does not establish 9 points of stock-selection alpha. A convenient weak benchmark exaggerates apparent success.

G2 | In-Sample vs. Out-of-Sample Evaluation

QF2-D6G2Q1 p. 675 / QF2-D6G2Q2 p. 675 / QF2-D6G2Q3 p. 676

Data that influence any design or parameter choice are in-sample for that choice. Freeze the candidate before an untouched test; a revision informed by the test requires new independent evidence. A predefined walk-forward retraining procedure can provide chronological forward evaluation if features are also point in time.

Watch the wording: Renaming a reused holdout does not restore its independence. Annual retraining does not imply independent yearly returns.

G3 | Return, Volatility, and Drawdown

QF2-D6G3Q1 p. 677 / QF2-D6G3Q2 p. 677 / QF2-D6G3Q3 p. 678

Compute cumulative return from ending relative to starting wealth, with cash flows handled consistently. Measure drawdown relative to a running peak, not only the original investment. Report return, volatility, and drawdown together when comparing different paths.

Watch the wording: Ending above the initial investment does not mean recovery to the prior peak. The recovery percentage after a loss exceeds the loss percentage because its base is smaller.

G4 | Risk-Adjusted Performance Metrics

QF2-D6G4Q1 p. 678 / QF2-D6G4Q2 p. 679 / QF2-D6G4Q3 p. 680

Sharpe compares excess return over the risk-free rate with total volatility. Sortino uses downside deviation below a stated target; information ratio uses active return and tracking error. Match the metric to the objective and inspect tails and drawdowns beyond a single ratio.

Watch the wording: Equal Sharpe and IR values can be a numerical coincidence, not equivalent definitions. Higher Sharpe does not guarantee a smaller maximum drawdown.

G5 | Turnover and Trading Behavior

QF2-D6G5Q1 p. 681 / QF2-D6G5Q2 p. 681 / QF2-D6G5Q3 p. 682

Turnover measures trading intensity and helps identify short holding periods or unnecessary churn. Frequent weak-signal reversals can spend gross alpha on spread, fees, and impact. Evaluate no-trade bands and slower updates by net performance and risk, not merely by reduced transaction count.

Watch the wording: Equal gross Sharpe does not imply equal net Sharpe. Lower turnover does not guarantee lower risk or better performance in every regime.

G6 | Performance Attribution

QF2-D6G6Q1 p. 683 / QF2-D6G6Q2 p. 683 / QF2-D6G6Q3 p. 684

Compare returns with exposures before calling outperformance alpha. Different long and short betas can leave market exposure even when net dollars are zero. Decompose sector allocation, selection, timing, and factor contributions under the stated attribution framework.

Watch the wording: A high-beta strategy beating a rising market may have no residual alpha. Aggregate active return alone does not reveal its sources.

G7 | Stability, Significance, and Robustness

QF2-D6G7Q1 p. 685 / QF2-D6G7Q2 p. 685 / QF2-D6G7Q3 p. 686

Examine subperiod and regime results and sensitivity to plausible costs and nearby parameters. Investigate concentration of profit in a few trades without assuming those trades are either repeatable or invalid. Distinguish statistical evidence of positive mean from economic value after operating and opportunity costs.

Watch the wording: Failure to reject zero does not prove the true mean is exactly zero. One strong regime or fragile parameter peak does not establish robustness.

Formulas and conditions

Cumulative return

$R = V_T / V_0 - 1$

No external cash flows in the pool examples; otherwise choose an appropriate cash-flow-adjusted method.

Drawdown and recovery

$Peak_t = \max_{s \le t} V_s$; $DD_t = 1 - V_t/Peak_t$; $MDD = \max_t DD_t$; recovery = $d/(1-d)$

Drawdown d is a positive loss fraction below 1; recovery is the gain needed to regain the same peak.

Sharpe ratio

$Sharpe = mean(r_p - r_f) / sd(r_p - r_f)$

Use consistent periods; for a constant risk-free return the denominator equals portfolio-return volatility. The pool's annual examples provide annual inputs directly.

Sortino ratio

$Sortino = mean(r_p - \tau) / \sqrt{mean(\min(r_p-\tau, 0)^2)}$

τ is a defined target; state the downside-deviation convention and use consistent frequency.

Information ratio

$IR = mean(r_p - r_b) / sd(r_p - r_b)$

Benchmark-relative numerator and denominator; changing the benchmark can change both.

Simplified beta attribution

$r_{beta} = r_f + \beta(r_m - r_f)$; $\alpha_{residual} = r_p - r_{beta}$

The exercise treats estimated beta and observed returns as given; residual return is not automatically proof of stable skill.

Work it through

Drawdown uses the peak

QF2-D6G3Q2 / pool p. 677

A portfolio goes from \$100 to \$120, falls to \$90, and ends at \$108.

Maximum observed drawdown = (120-90)/120 = 25%. Ending cumulative return = 108/100-1 = 8%. Recovery from \$90 to \$120 requires 30/90 = 33.3%.

Result: The portfolio finishes profitable but remains 10% below its \$120 peak.

Higher return, lower Sharpe

QF2-D6G4Q1 / pool p. 678

Risk-free rate is 2%. A earns 10% with 16% volatility; B earns 8% with 8% volatility.

Sharpe_A = (10-2)/16 = 0.50. Sharpe_B = (8-2)/8 = 0.75.

Result: B has the higher Sharpe despite A's higher raw return.

Beta can explain outperformance

QF2-D6G6Q1 / pool p. 683

A strategy earns 14%, cash earns 2%, the market earns 10%, and estimated beta is 1.5.

Beta-implied return = 2% + 1.5×(10%-2%) = 14%. Residual under the stated model = 14%-14% = 0%.

Result: The 4-point advantage over the market does not establish alpha.

QF2-D7 / QUANTITATIVE FINANCE / 21 QUESTIONS

Trading Frictions, Capacity, and Implementability

Core understanding

Costs extend beyond commissions. Explicit commissions and fees are billed directly. Implicit costs include crossing the spread, slippage relative to a specified benchmark, market impact, and missed opportunities from incomplete execution. Identify the reference price before computing a cost. A later favorable price movement does not erase the cost of implementing the original decision, and overlapping cost labels should not be counted twice.

Turnover turns small costs into large drag. The expected net edge is the expected gross edge minus the costs needed to enter, hold, and exit. A small positive forecast can be uneconomic when round-trip costs exceed it. Turnover multiplies trading cost per dollar into portfolio-level drag. Thresholds, no-trade bands, and slower rebalancing may improve net results while sacrificing some gross alpha.

Liquidity and capacity depend on size. A quote applies only to the depth available at that price. Large orders can consume several book levels and move prices; average daily volume does not guarantee a fill. Scaling capital raises dollar trading requirements without creating opportunities. Capacity depends on the signal's horizon, turnover, market depth, impact, and the quality of marginal trades, especially under stress.

Funding and crowding change feasibility. Short sales require available borrow, may incur changing fees, and can be recalled; margin increases can force deleveraging. Similar investors can enter or exit the same positions together, reducing alpha and worsening liquidity. Short squeezes and synchronized unwinds can make normal-period cost assumptions unreliable even when the original predictive relationship was real.

Every source group

G1 | Explicit and Implicit Trading Costs

QF2-D7G1Q1 p. 689 / QF2-D7G1Q2 p. 689 / QF2-D7G1Q3 p. 690

Separate directly billed fees from spread, slippage, impact, and non-execution opportunity costs. For a buy, compare fill with the stated arrival reference; decompose spread and additional slippage without double counting. Post-trade reversion can be consistent with temporary pressure but does not prove that the full order was executable at the later price earlier.

Watch the wording: Zero commission does not mean zero trading cost. A better ending midpoint is not a replacement for the decision-time benchmark.

G2 | Turnover and Cost Drag

QF2-D7G2Q1 p. 691 / QF2-D7G2Q2 p. 692 / QF2-D7G2Q3 p. 692

Apply the explicitly stated turnover convention to compute total dollars traded. Compare forecasted gross edge with the relevant one-way or round-trip cost before trading. Higher thresholds and slower rebalancing can reduce churn while also missing or delaying opportunities.

Watch the wording: Higher gross alpha can coexist with lower net alpha. A forecast above a cost threshold does not guarantee each trade profits.

G3 | Liquidity and Tradability

QF2-D7G3Q1 p. 693 / QF2-D7G3Q2 p. 694 / QF2-D7G3Q3 p. 694

Compare order size with market volume and book depth, not only the quoted spread. A market order can consume several prices, making its average fill worse than the best quote. Stress liquidity assumptions because spreads, depth, and exit time can deteriorate together.

Watch the wording: Average daily volume is a scale estimate, not guaranteed executable liquidity. Equal top-of-book spreads do not imply equal impact for large trades.

G4 | Market Impact and Order Size

QF2-D7G4Q1 p. 695 / QF2-D7G4Q2 p. 695 / QF2-D7G4Q3 p. 696

Participation measures the order's size relative to the stated market volume. Larger and faster orders can increase impact, potentially nonlinearly. Slower schedules can reduce immediate pressure but add timing risk and lose decaying alpha. A post-order price reversal is consistent with temporary impact; persistent price movement alone does not prove permanent causal impact.

Watch the wording: Lowest estimated spread and impact does not necessarily mean lowest total shortfall. Peak price displacement is not automatically the permanent-impact component.

G5 | Capacity and Scaling

QF2-D7G5Q1 p. 697 / QF2-D7G5Q2 p. 697 / QF2-D7G5Q3 p. 698

More capital requires more traded dollars for a fixed turnover rate, without automatically expanding opportunities. Net return can fall because gross alpha dilutes and costs rise at larger scale. Deeper markets, lower turnover, or longer horizons may expand capacity but can change forecast strength and risk.

Watch the wording: Small-scale percentage returns are not guaranteed at much larger capital. Largest gross dollar profit does not identify the best net deployment.

G6 | Shorting, Borrowing, and Funding Constraints

QF2-D7G6Q1 p. 699 / QF2-D7G6Q2 p. 699 / QF2-D7G6Q3 p. 700

Model borrow availability, variable borrow fees, recall risk, financing, and margin constraints. Gross exposure adds absolute long and short positions; net exposure subtracts shorts from longs. Rising short prices or margin requirements can force covering or deleveraging despite unchanged forecasts.

Watch the wording: Historical price data alone do not establish that a stock was shortable. An uncovered short stock position can lose more than its initial market value.

G7 | Crowding and Implementation Risk

QF2-D7G7Q1 p. 701 / QF2-D7G7Q2 p. 702 / QF2-D7G7Q3 p. 702

Shared signals can produce overlapping positions and synchronized sales of longs or covering of shorts. Competitors may capture a predicted move before the assumed backtest fill. Assess aggregate participation and stressed joint exits, not only an individual strategy in calm markets.

Watch the wording: Public data do not guarantee equal execution access or latency. Crowding can affect liquidity and drawdown as well as average alpha.

Formulas and conditions

Basis points and net edge

1 bp = 0.0001 = 0.01%; expected net edge ≈ gross edge - all relevant costs

Match horizon and one-way versus round-trip cost conventions.

Turnover cost drag

Traded dollars = turnover × NAV; cost = traded dollars × cost rate; drag = turnover × cost rate

Here turnover means total dollar volume traded divided by portfolio value, as explicitly defined in D7G2Q1.

Buy execution cost relative to arrival

Cost = Q(P_fill - P_arrival) + explicit fees

Completed buy with average fill; opportunity cost for unfilled shares is additional. Spread and impact may already be included in the price difference.

Participation rate

Participation = order volume / stated market volume

Use matching shares or dollar units and the same interval; ADV is a forecast proxy, not realized depth.

Weighted average fill

$P_{fill} = \Sigma(q_j P_j) / \Sigma q_j$

Weights are actually executed shares at each price level.

Gross leverage

Gross leverage = (long market value + |short market value|) / equity

Net exposure uses long minus absolute short; neither gross nor net alone measures portfolio covariance risk.

Work it through

Spread, slippage, and explicit fees

QF2-D7G1Q1 / pool p. 689

Buy 1,000 shares with bid $49.95, ask $50.05, midpoint $50.00, fill $50.07, and $5 commission.

Spread is $0.10. Crossing from midpoint to ask costs $0.05×1,000 = $50. Additional slippage beyond ask costs $0.02×1,000 = $20. Add $5 commission.

Result: Total cost versus arrival midpoint = $75; do not add the full spread again.

Turnover scales the same per-dollar fee

QF2-D7G2Q1 / pool p. 691

Two $10 million funds trade 20% and 200% of NAV monthly. One-way cost is 10 bps under the stated total-dollar-volume convention.

Traded amounts are $2 million and $20 million. At 0.001 cost per dollar, monthly costs are $2,000 and $20,000, or 0.02% and 0.20% of NAV.

Result: The high-turnover strategy spends ten times more on this cost component.

Walk the book

QF2-D7G3Q2 / pool p. 694

A 3,000-share buy sees 500 shares at $20.10, 1,000 at $20.30, and 2,000 at $20.60, with no new liquidity.

It consumes 500 + 1,000 + 1,500 shares. Dollar cost = 500×20.10 + 1,000×20.30 + 1,500×20.60 = $61,250. Average fill = 61,250/3,000 ≈ $20.4167.

Result: The best ask is not the executable average price for the entire order.

Capacity changes net value

QF2-D7G5Q2 / pool p. 697

At $10m, $50m, and $200m, estimated gross alpha is 6%, 5%, and 3%, with costs of 1%, 2%, and 4%.

Net alpha is 5%, 3%, and -1%. Approximate net alpha dollars are $0.5m, $1.5m, and -$2m respectively under these estimates.

Result: Greater capital can increase net dollars initially while reducing percentage return; excessive scale can destroy value.

QF2-D8 / QUANTITATIVE FINANCE / 21 QUESTIONS

Execution Algorithms and Order Book Analytics

Core understanding

Choose the benchmark that matches the task. Arrival price evaluates the cost of implementing a decision from its starting point. VWAP weights market prices by traded volume; TWAP averages across time intervals; the close can matter for a closing mandate. A fill may beat one benchmark and lag another. Evaluate the full parent decision, including unfilled quantity, rather than rewarding only favorable completed shares.

Price improvement competes with completion. An aggressive buy normally consumes asks; an aggressive sell consumes bids. Passive limits may save the spread but can wait, partially fill, or miss the trade. Limit price, queue position, displayed depth, cancellations, and urgency influence execution. A cheaper quoted limit is not necessarily a cheaper economic outcome when the signal disappears or the market moves away.

Scheduling is a risk tradeoff. Fast execution reduces exposure of the remaining order to price movement and signal decay, but can increase impact. Slower execution spreads demand over time while accepting waiting risk. VWAP follows a forecast volume shape, TWAP allocates through equal time intervals, and participation strategies react to observed volume. Unexpected volume and liquidity changes can require a schedule update.

Displayed liquidity and filled orders need context. Order-book imbalance describes visible demand and supply at an instant, not certain price direction or guaranteed liquidity. Quote persistence, deeper levels, cancellations, and latency matter. Passive orders can fill disproportionately just before adverse moves, and predictable child orders can reveal remaining demand. Evaluate post-fill markouts, completion, impact, and leakage together.

Every source group

G1 | Execution Objectives and Benchmarks

QF2-D8G1Q1 p. 706 / QF2-D8G1Q2 p. 706 / QF2-D8G1Q3 p. 707

Select arrival, VWAP, TWAP, or closing benchmarks according to the execution objective. For completed buys, arrival shortfall includes quantity times fill-minus-arrival plus explicit costs. Include the opportunity cost of unfilled shares when evaluating the whole parent decision.

Watch the wording: A later profitable market move does not erase decision-time execution cost. Beating VWAP on average does not prove every child order was favorable.

G2 | Aggressive vs. Passive Execution

QF2-D8G2Q1 p. 708 / QF2-D8G2Q2 p. 708 / QF2-D8G2Q3 p. 709

Marketable buys normally interact with asks; marketable sells normally interact with bids, subject to available depth. Passive limits can improve price if filled while exposing the remaining quantity to delay and non-completion. Choose aggressiveness using urgency and opportunity cost as well as the quoted spread.

Watch the wording: Posting at the best bid does not guarantee a buy fill. A favorable partial fill does not establish superior execution for the entire decision.

G3 | Order Scheduling Over Time

QF2-D8G3Q1 p. 710 / QF2-D8G3Q2 p. 710 / QF2-D8G3Q3 p. 711

Faster schedules usually reduce unexecuted exposure time but can demand more liquidity and impact. Front-loading executes more before a specified event; it reduces remaining exposure to that event without guaranteeing a better price. Update scheduling as volatility, depth, urgency, and the signal’s decay change.

Watch the wording: Slower execution does not always minimize total implementation shortfall. A deadline or risk-limit change can justify higher immediate impact.

G4 | VWAP, TWAP, and Participation Strategies

QF2-D8G4Q1 p. 712 / QF2-D8G4Q2 p. 712 / QF2-D8G4Q3 p. 713

A VWAP-style schedule follows forecast volume fractions; TWAP allocates equally to equal time intervals. A percentage-of-volume strategy targets a stated fraction of observed volume and adapts its pace to activity. Forecast-volume errors can leave too much quantity for illiquid later intervals; dynamic updates may help.

Watch the wording: Fixed participation does not guarantee completion by a clock-time deadline. TWAP does not require an intraday market-volume forecast by definition.

G5 | Limit Order Placement and Fill Probability

QF2-D8G5Q1 p. 714 / QF2-D8G5Q2 p. 714 / QF2-D8G5Q3 p. 715

A less competitive limit generally trades price improvement for lower immediate fill probability. Under price-time priority, earlier same-price orders usually have better queue positions, subject to cancellations. After partial fills, re-evaluate remaining quantity, urgency, price movement, and the queue effect of repricing.

Watch the wording: Equal limit prices do not imply equal fill probabilities. A partial fill does not settle the best action for the remaining order.

G6 | Order Book Imbalance and Short-Term Signals

QF2-D8G6Q1 p. 716 / QF2-D8G6Q2 p. 716 / QF2-D8G6Q3 p. 717

Compute the defined imbalance from bid and ask depth at the stated book levels. Positive best-level imbalance can indicate upward pressure under some conditions but remains a noisy, short-lived signal. Check quote persistence, cancellations, deeper liquidity, execution latency, and costs before using a snapshot signal.

Watch the wording: Large displayed bids need not be reliable executable support. A suspicious cancellation pattern alone does not prove manipulation.

G7 | Adverse Selection and Execution Quality

QF2-D8G7Q1 p. 718 / QF2-D8G7Q2 p. 718 / QF2-D8G7Q3 p. 719

Adverse selection occurs when fills tend to arrive just before price moves against the passive trader. Assess post-fill price movement and completion alongside spread savings and fill rate. Predictable child-order patterns can leak remaining demand; changing patterns may reduce but not eliminate leakage.

Watch the wording: A passive fill does not automatically earn the spread as profit. Splitting a parent order does not automatically hide its intent.

Formulas and conditions

VWAP and TWAP

$VWAP = \Sigma(P_j V_j)/\Sigma V_j$; $TWAP = (1/m)\Sigma P_j$

VWAP weights actual market volume; this TWAP expression uses equal-duration intervals and defined representative prices.

Full-decision buy shortfall

$IS = Q_f(P_{fill}-P_0) + Q_u(P_{end}-P_0) + fees$

P_0 is decision/arrival reference; Q_f and Q_u are filled and unfilled shares. P_{end} values the missed opportunity at the chosen evaluation endpoint.

Participation target

$q_j = p \times V_j$

Simplified percentage-of-observed-volume rule with the question's stated market-volume convention and no other limits.

Order-book imbalance

$I = (D_{bid} - D_{ask}) / (D_{bid} + D_{ask})$

Nonzero total displayed depth; range [-1,1]. Scope can be best level or explicitly aggregated levels.

Signed execution cost and markout

Cost/share $= s(P_{fill}-P_{ref})$; markout $= s(P_{later}-P_{fill})$, $s = +1$ for buy and -1 for sell

Positive cost is unfavorable; positive markout is favorable. Specify later-price horizon and use comparable references.

Work it through

Arrival shortfall

QF2-D8G1Q1 / pool p. 706

Buy 20,000 shares from a \$40.00 arrival midpoint, fill at \$40.12, and pay \$200 commission.

Price shortfall = 20,000×0.12 = \$2,400; including fees = \$2,600. Relative to \$800,000 arrival notional, cost is 0.325% = 32.5 bps.

Result: A \$40.30 later close does not change this \$2,600 execution cost.

Account for unfilled shares

QF2-D8G1Q3 / pool p. 707

The decision is to buy 100,000 shares at arrival \$60.00. Fill 70,000 at \$60.10; 30,000 remain unfilled when price closes \$60.50. Ignore fees.

Filled shortfall = 70,000×0.10 = \$7,000. Missed-opportunity cost = 30,000×0.50 = \$15,000. Full-decision shortfall = \$22,000. The fills alone are \$0.40 below the close.

Result: Favorable filled-share comparison with the close can coexist with substantial total implementation shortfall.

Participation follows observed volume

QF2-D8G4Q2 / pool p. 712

A 10% participation rule sees intervals with 100,000, 300,000, and 50,000 market shares.

Targets are 10,000, 30,000, and 5,000 shares, totaling 45,000. If the parent order is larger, it remains incomplete absent later volume or changed instructions.

Result: A percentage-of-volume target does not promise completion by a fixed deadline.

Read imbalance correctly

QF2-D8G6Q1 / pool p. 716

Displayed best-bid depth is 80,000 shares and best-ask depth is 20,000.

I = (80,000-20,000)/(80,000+20,000) = 0.60. There is four times as much displayed bid as ask depth at this instant.

Result: Positive 0.60 is a snapshot feature, not a 60% guaranteed price rise or a certainty about the next tick.

QF2-D9 / QUANTITATIVE FINANCE / 21 QUESTIONS

Options and Derivative Payoffs

Core understanding

Rights and obligations have different signs. A call holder has the right to buy at the strike; a put holder has the right to sell. At expiration, the long holder exercises only when the payoff is positive. The writer has the corresponding obligation and opposite option payoff. A long option's payoff is never negative, while its profit can be negative because the buyer paid a premium.

Keep payoff, profit, and moneyness separate. Call payoff is max(S_T-K,0); put payoff is max(K-S_T,0). Buyer profit subtracts the premium and writer profit adds it to the opposite payoff. Moneyness compares spot with strike and ignores the premium; therefore an in-the-money option can still lose money. State whether values are per share or multiplied by the contract size.

Before expiry, possibility has value. Intrinsic value is the nonnegative immediate-exercise amount. An out-of-the-money option can still have value because time remains for favorable terminal outcomes. Greater volatility can increase a standard long option's value through its asymmetric payoff, with other pricing inputs fixed. Time decay is a conditional comparison, not a guarantee of a daily price decline while markets move.

Hedges reshape, rather than erase, risk. A stock plus protective put has a terminal floor before premium and retains upside, but its insurance costs money. A covered call receives premium while capping stock upside above the strike and retaining substantial downside. Long calls, puts, straddles, stock, and uncovered writers have different piecewise exposures; the same bullish direction does not make them fixed-leverage equivalents.

Every source group

G1 | Call Option Payoffs

QF2-D9G1Q1 p. 723 / QF2-D9G1Q2 p. 723 / QF2-D9G1Q3 p. 724

A long call pays max(S_T-K,0) at expiration and gives a right rather than an obligation to buy. A call at or below strike expires with zero exercise payoff; premium can still create a loss. An uncovered short call has the opposite payoff and potentially unbounded loss as the stock price rises.

Watch the wording: Call payoff is not negative when the stock finishes below strike. The short call's loss is not capped at the strike.

G2 | Put Option Payoffs

QF2-D9G2Q1 p. 725 / QF2-D9G2Q2 p. 725 / QF2-D9G2Q3 p. 726

A long put pays max(K-S_T,0) and can be bought for speculation without owning stock. For a nonnegative stock price, maximum long-put payoff is K; the writer has the opposite payoff. Short-put profit equals received premium minus payoff obligation, so maximum loss is K minus premium under the stated assumptions.

Watch the wording: A put is valuable at expiration when the stock is below, not above, its strike. A bounded short-put loss can still be substantial; it differs from uncovered short-call loss.

G3 | Premium, Payoff, Profit, and Breakeven

QF2-D9G3Q1 p. 727 / QF2-D9G3Q2 p. 727 / QF2-D9G3Q3 p. 728

Buyer profit subtracts premium from expiration payoff; writer profit is the exact opposite before other costs. Call breakeven is K+C; put breakeven is K-P at expiration under the simplified undiscounted calculation. A positive exercise payoff can be smaller than the paid premium and leave a net loss.

Watch the wording: Received premium is not the writer's final profit when exercise creates an obligation. At-the-money expiration does not reimburse the buyer's premium.

G4 | Moneyness and Intrinsic Value

QF2-D9G4Q1 p. 728 / QF2-D9G4Q2 p. 729 / QF2-D9G4Q3 p. 730

Calls are in the money when S>K; puts are in the money when S<K; equality is at the money. Intrinsic value is floored at zero and excludes premium. Stock-price changes can change moneyness even when the strike is fixed.

Watch the wording: Out-of-the-money intrinsic value is zero, not negative. A call and put with the same strike cannot both be strictly in the money at the same spot price.

G5 | Time Value and Volatility Intuition

QF2-D9G5Q1 p. 730 / QF2-D9G5Q2 p. 731 / QF2-D9G5Q3 p. 732

An out-of-the-money option can have positive pre-expiry market value despite zero intrinsic value. Under the standard comparison with other inputs fixed, greater volatility can raise long-call value because upside payoff is retained and downside payoff is floored. For the out-of-the-money call in the question, less remaining time generally reduces time value; no time value remains at expiration.

Watch the wording: A positive option price with zero intrinsic value is not necessarily a pricing error. Time decay does not guarantee a lower observed price every day when spot and volatility also change.

G6 | Hedging with Options

QF2-D9G6Q1 p. 732 / QF2-D9G6Q2 p. 733 / QF2-D9G6Q3 p. 734

A protective put creates a terminal stock-value floor at its strike before the insurance premium. A covered call owns stock against the written call, receives premium, and gives up gains above the strike. Strike choice trades protection against cost; a hedge on a different underlying can leave basis risk.

Watch the wording: Protection need not prevent a loss relative to original purchase price plus premium. Covered calls do not retain unlimited stock upside or eliminate stock downside.

G7 | Nonlinear and Asymmetric Payoffs

QF2-D9G7Q1 p. 734 / QF2-D9G7Q2 p. 735 / QF2-D9G7Q3 p. 736

Long stock has linear terminal exposure; option payoffs change slope at their strikes. A long call loses at most premium, while an uncovered short call earns at most premium and can lose without a fixed upper bound. A same-strike long call plus long put benefits from sufficiently large moves in either direction, after covering both premiums.

Watch the wording: A call is not a fixed multiple of stock across terminal prices. Below strike, further stock declines do not further reduce a long call's zero expiration payoff.

Formulas and conditions

Long and short call

Call payoff = max(S_T-K,0); long profit = max(S_T-K,0)-C; short profit = C-max(S_T-K,0)

Per share, held to expiration; ignore other costs and the financing value of the premium, as in the pool.

Long and short put

Put payoff = max(K-S_T,0); long profit = max(K-S_T,0)-P; short profit = P-max(K-S_T,0)

Per share at expiration; assume $S_T \geq 0$ for bounded maximum put payoff.

Expiration breakeven

Long call: S_T = K+C; long put: S_T = K-P

Premiums C and P are per share; ignores commissions, dividends, and financing.

Contract scaling

Contract payoff or profit = per-share amount × contract multiplier

Use the explicitly stated multiplier; the pool examples specify 100 shares when needed.

Protective put

Terminal value = S_T + max(K-S_T,0) = max(S_T,K); profit = max(S_T,K)-S_0-P

One put unit per share of stock, held to expiry; ignore dividends, financing, and other costs.

Covered call

Profit = S_T-S_0 + C-max(S_T-K,0) = min(S_T,K)-S_0+C

One written call unit per owned share; upside caps at K-S_0+C and substantial stock downside remains.

Long straddle

Payoff = |S_T-K|; profit = |S_T-K|-(C+P); breakevens = K ± (C+P)

Same underlying, strike, expiration, and units; ignore financing and other costs; a lower breakeven must be feasible for $S_T \geq 0$.

Work it through

Positive payoff, negative profit

QF2-D9G3Q1 / pool p. 727

Pay $6 for a $100-strike call. At expiration the stock is $103.

Call payoff = max(103-100,0) = $3. Profit = 3-6 = -$3 per share. Breakeven is 100+6 = $106.

Result: In the money does not mean profitable.

A short put's bounded maximum loss

QF2-D9G2Q3 / pool p. 726

Write a $60-strike put for a $4 premium and hold through expiration.

At or above $60, payoff obligation is zero and profit is $4. At stock price zero, obligation is $60 and profit is 4-60 = -$56. Breakeven is $56.

Result: Maximum profit is $4 and maximum loss is $56 per share, not $64.

Insurance floor versus total profit

QF2-D9G6Q1 / pool p. 732

Buy stock at $80 and a $75-strike protective put for $3. Stock finishes at $50.

Stock loss = 50-80 = -$30. Put payoff = 75-50 = $25; put net result = 25-3 = $22. Combined profit = -$8. Terminal stock-plus-put value is $75 against total cost $83.

Result: The hedge floors terminal value at $75 but does not guarantee no loss on the $83 outlay.

A straddle needs a large enough move

QF2-D9G7Q2 / pool p. 735

Buy a $100-strike call and put for $6 each.

Total premium is $12. At $100, profit = -$12. At $125, payoff = $25 and profit = $13. At $80, payoff = $20 and profit = $8. Breakevens are $88 and $112.

Result: The expiration profit is V-shaped with maximum loss $12 at the strike.

QF2-D10 / QUANTITATIVE FINANCE / 21 QUESTIONS

AI-Enabled Investment Processes and Decision Support

Core understanding

Evaluate the entire investment workflow. AI can summarize sources, forecast returns, propose weights, execute orders, monitor risk, and explain decisions. Each stage has its own objectives and failure modes. Accurate research summaries do not prove predictive skill; predictive skill does not guarantee feasible allocation or profitable execution. Client explanations must remain grounded in the reasons and evidence actually used.

Calibrate autonomy to the decision. Decision support, bounded automation, and approval-based escalation can coexist. Define who has authority, what limits apply, and when conflicting feeds or unfamiliar conditions trigger a pause or review. Human contextual knowledge can complement broad AI screening, but review is meaningful only when it can challenge assumptions. Neither human involvement nor a high confidence score guarantees correctness.

Integrate evidence with provenance. Knowledge graphs represent entities and relationships so an event can be traced through suppliers, people, and securities. Symbolic rules can enforce explicit restrictions while statistical models estimate forecasts. Correct entity resolution, recency, timing, and materiality remain necessary. Structured numbers and unstructured text can complement or contradict each other; more modalities do not repair corrupted alignment.

Govern behavior as markets change. A readable explanation may not faithfully reflect a model's actual drivers. Versioned decisions, evidence, approvals, and justified overrides support accountability. Monitor data freshness and cross-feed agreement as well as input distributions and forecast performance. Fixed code can drift as language, competition, or market structure changes, and shared models can synchronize selling across otherwise independent firms.

Every source group

G1 | AI Across the Investment Workflow

QF2-D10G1Q1 p. 739 / QF2-D10G1Q2 p. 740 / QF2-D10G1Q3 p. 740

Evaluate research, forecasting, construction, execution, monitoring, and communication against their own objectives and risks. Assess net portfolio results because accurate forecasts can be overwhelmed by illiquid allocations and costly execution. Ground generated explanations in approved decision records and check the interfaces between stages.

Watch the wording: Success at one stage does not validate the entire platform. Approved trades do not authorize invented explanations of their rationale.

G2 | Decision Support vs. Autonomous Decision-Making

QF2-D10G2Q1 p. 741 / QF2-D10G2Q2 p. 741 / QF2-D10G2Q3 p. 742

Autonomy can be bounded differently for ordinary low-impact actions and novel or high-impact decisions. Predefined limits, escalation rules, monitoring, and safe fallbacks support controlled automation. Conflicting feeds or out-of-distribution conditions can warrant pausing even when model confidence is high.

Watch the wording: Strong backtest performance alone does not establish readiness for full autonomy. Partial autonomy is coherent; automation does not eliminate model risk.

G3 | Human Intelligence Plus Artificial Intelligence

QF2-D10G3Q1 p. 743 / QF2-D10G3Q2 p. 743 / QF2-D10G3Q3 p. 744

Combine AI's scale with human domain knowledge and investigation of anomalous or material cases. Interpret confidence by its definition and calibration across relevant regimes. Document disagreement and overrides, then assess the combined process rather than assuming either component is always right.

Watch the wording: A 92% score does not automatically mean a 92% probability of profit. Rubber-stamping a score threshold contributes little independent review.

G4 | Knowledge Graphs and Symbolic Representation

QF2-D10G4Q1 p. 745 / QF2-D10G4Q2 p. 745 / QF2-D10G4Q3 p. 746

Graphs distinguish entities and relationship types and can support multi-step exposure queries. Statistical forecasts and symbolic restrictions can serve complementary roles in one process. Check entity resolution, relationship recency, and economic materiality before propagating an event into signals.

Watch the wording: A connection does not establish equal impact or causal return predictiveness. A strong forecast does not automatically override a restricted-list rule.

G5 | Structured and Unstructured Information Integration

QF2-D10G5Q1 p. 747 / QF2-D10G5Q2 p. 748 / QF2-D10G5Q3 p. 748

Align every modality to the correct entity, fiscal period, and actual information-availability time.

Maintain historical data vintages when vendors revise series; inspect coverage and nonrandom missingness. Weigh conflicting sources using provenance, purpose, and context rather than automatically averaging them.

Watch the wording: Accurate numeric fields do not neutralize mislabeled text features. Alternative data do not eliminate bias, collection problems, or provenance requirements.

G6 | Explainability, Governance, and Oversight

QF2-D10G6Q1 p. 749 / QF2-D10G6Q2 p. 750 / QF2-D10G6Q3 p. 750

Validate whether explanations faithfully reflect model behavior as well as whether they read clearly. Material changes require review of intended use, assumptions, data, controls, and failures, not only validation Sharpe. Keep versioned audit trails and contemporaneous records of override evidence, authority, and timing.

Watch the wording: A plausible story does not prove what drove a recommendation. A profitable override does not by itself validate the override process.

G7 | Operational, Model, and Systemic Risk

QF2-D10G7Q1 p. 751 / QF2-D10G7Q2 p. 752 / QF2-D10G7Q3 p. 752

Monitor freshness and independent feed agreement because schema-valid values can still be stale. Detect changes in input distributions and input-outcome relationships even if code remains unchanged. Stress shared-model and vendor dependencies because synchronized risk controls can amplify market pressure.

Watch the wording: Unchanged software does not rule out model drift. Many securities or separate legal owners do not guarantee independence from systemic AI-driven flows.

Formulas and conditions

Net workflow value

Expected net alpha ≈ gross forecast alpha - allocation and execution costs

Use a common horizon and avoid overlapping cost categories; operational and forecast errors can add further losses.

Confidence calibration check

Observed event rate in score bin B = count(successes in B) / count(predictions in B)

Only compare with nominal probabilities when the score explicitly predicts that event; evaluate sample size and regime.

Information-availability constraint

feature availability time ≤ decision time < evaluated future outcome interval

Use actual release and revision timestamps, not only fiscal-quarter labels or the latest reconstructed history.

Work it through

Forecast quality is not net performance

QF2-D10G1Q2 / pool p. 740

An AI system has 6% annual gross alpha, while estimated costs and impact are 8%.

Estimated net alpha = 6%-8% = -2%. Examine concentration, liquidity limits, trade urgency, and smaller or slower execution while checking alpha decay.

Result: Better forecasting alone has not produced an investable implementation.

A confidence label needs a definition

QF2-D10G3Q3 / pool p. 744

A committee automatically approves scores above 90%; the model reports 92%.

The score's target and calibration must be known before reading it as a probability. As an illustrative audit, if 100 comparable "92%" profit-probability forecasts produce only 60 profitable trades, the observed rate is 60%, with sampling and regime uncertainty to investigate.

Result: A label alone cannot replace evidence and contextual review; the 100-trade audit is illustrative, not a reported pool result.

Trace a graph without inventing causality

QF2-D10G4Q1 / pool p. 745

A factory shutdown affects a private supplier linked to several public holdings in a knowledge graph.

Traverse supplier-to-company-to-security links, verify that relationships are current, and assess dependence, substitutability, materiality, and timing for each holding. The graph identifies investigation candidates, not identical impacts.

Result: Structured connections organize evidence; they do not by themselves prove a trade or its magnitude.

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Mixed self-test | Questions 1-10

Answer without notes. Show your calculation and state the controlling assumption. The separate answer section explains the result.

01 / ML2-D1

A charging controller sees battery 30%, chooses recharge, then receives reward -2 for delay and later +12 for completing a delivery. Identify state, action, and reward, and explain why the -2 does not alone condemn the action.

02 / ML2-D2

Action A pays 3 now then leads to a state valued at 4. Action B pays 0 now and leads to value 12 with probability 0.25 or value 2 with probability 0.75. With $\gamma = 0.8$, which has the higher one-step value?

03 / ML2-D3

Old Q = 4, reward = -1, max successor Q = 6, $\alpha = 0.25$, $\gamma = 0.5$. Compute the nonterminal update. What if the same reward instead ends the episode?

04 / ML2-D4

There are four actions, two tied for largest Q. Epsilon is 0.20; exploration is uniform over all four, and the greedy branch breaks its tie uniformly. What is the probability of each tied action and each other action?

05 / ML2-D5

A Dyna agent observes 80 real transitions and performs 15 planning updates after each. A modeled pair has 12 visits to A and 8 to B. How many Q-updates occur, and what model probabilities should be used?

06 / ML2-D6

Three indicators have 4, 5, and 6 bins; holdings has three categories and there are three actions. How large is the nominal Q-table, and what does doubling only the first indicator’s bins do?

07 / ML2-D7

A portfolio is short 400 shares and requests a long target of 400 shares. Starting value is $50,000; ending value before a $15 commission is $50,200. Find the order, net dollar reward, and net percentage return.

08 / ML2-D8

An RL system beats a tree in a later period, but RL sees holdings, has zero impact costs, and was selected from 30 variants using that period. What must change before attributing the edge to the algorithm?

09 / ML2-D9

A dense network has two inputs, three hidden units, one output, and a bias per unit. How many parameters does it have? With 1,200 training rows and batches of 80, how many updates occur in four epochs?

10 / ML2-D10

A recurrent model forecasts a company from past filings, but carries its hidden state over from an unrelated company; an LLM then cites a genuine filing for a wrong percentage. Identify the two independent checks.

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Mixed self-test | Questions 11-20

Answer without notes. Show your calculation and state the controlling assumption. The separate answer section explains the result.

11 / QF2-D1

A price rises 20% and then falls 20%. What is the cumulative return, and would summing log returns change the ending wealth?

12 / QF2-D2

A band uses $\mu = 50$, $\sigma = 3$, and $k = 2$. Price is 44. Compute the lower band and z-score; does the touch guarantee a rebound?

13 / QF2-D3

A signal is strongest on day 1 and has no detectable IC by day 10. Why might daily rebalancing still be inferior to a slower or staggered rule?

14 / QF2-D4

A strategy has IC 0.06 and breadth 100. Expansion halves IC and quadruples effective breadth. What happens to predicted IR?

15 / QF2-D5

A portfolio has 160% long exposure and 60% short exposure. Compute gross and net exposure. Does net 100% establish compliance with a 150% gross cap?

16 / QF2-D6

A strategy earns 1.5 percentage points more than its passive benchmark before costs and requires 2.0 percentage points of extra costs. What is its net active return?

17 / QF2-D7

A trade has 8 bps of expected gross edge and 10 bps of round-trip costs. Another rule trades less, with 7% annual gross alpha and 3% costs. Compute both net values.

18 / QF2-D8

A passive buy may save $0.04 relative to crossing, but filled passive buys have a subsequent average adverse move of $0.07. Does the spread saving alone establish better execution?

19 / QF2-D9

An investor owns stock bought at $100 and writes a $110 call for $4. What are profits if stock ends at $105, $130, or $0?

20 / QF2-D10

A price feed is 20 minutes stale but passes its numeric schema checks, and the model reports high confidence. Which controls address the actual failure?

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Self-test answers | 1-10

Compare the reasoning as well as the number. Check the horizon, denominator, information timing, and selection convention.

01 / ML2-D1

Battery level is part of state; recharge is the chosen action; -2 and +12 are rewards from outcomes. The recharge changes future feasibility, so evaluate its cumulative consequences, not only the immediate delay cost.

02 / ML2-D2

A: 3 + 0.8×4 = 6.2. B successor expectation: 0.25×12 + 0.75×2 = 4.5; B total = 0.8×4.5 = 3.6. A has the greater value under these continuation estimates.

03 / ML2-D3

Nonterminal target = -1+0.5×6 = 2; new Q = 4+0.25×(2-4) = 3.5. If terminal, target = -1 and new Q = 4+0.25×(-1-4) = 2.75. The reset state supplies no future value to a terminal transition.

04 / ML2-D4

Each tied action gets 0.80/2 + 0.20/4 = 0.45. Each nongreedy action gets 0.20/4 = 0.05. The four probabilities sum to 1. Setting epsilon to zero would still leave random tie-breaking.

05 / ML2-D5

There are 80 direct and 80×15 = 1,200 planning updates, totaling 1,280 Q-updates but only 80 real transitions. For the pair, T(A)=12/20=0.6 and T(B)=8/20=0.4; repeated planning does not add independent observations to those counts.

06 / ML2-D6

States = 4×5×6×3 = 360; entries = 360×3 = 1,080. Doubling only the first dimension gives 720 states and 2,160 entries. Finer detail doubles nominal data requirements for equal visits per entry but does not guarantee that all combinations are reachable or useful.

07 / ML2-D7

Buy 400-(-400)=800 shares. Net ending value is $50,185; reward = $185. Return = $185/$50,000 = 0.0037 = 0.37%, assuming no other costs or external cash flows.

08 / ML2-D8

Align information access across the complete pipelines, execution and cost assumptions, and capital/constraints. Treat the period used to select among 30 variants as development data and evaluate the finalized systems on an untouched later holdout with common metrics and benchmarks.

09 / ML2-D9

Parameters = 2×3+3 + 3×1+1 = 13. Batches per epoch = 1,200/80 = 15; four epochs give 60 updates if each batch triggers one update. Neither the parameter count nor the update count proves generalization.

10 / ML2-D10

Reset or deliberately initialize recurrent state at independent company boundaries so the intended sequence determines the forecast. Separately verify the generated percentage against the cited source and recompute it; a genuine citation does not guarantee that the claim is supported.

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Self-test answers | 11-20

Compare the reasoning as well as the number. Check the horizon, denominator, information timing, and selection convention.

11 / QF2-D1

(1.20 × 0.80) - 1 = -4%; $100 becomes $96. Log returns sum to ln(0.96), which converts back to -4%. The representation does not change wealth.

12 / QF2-D2

Lower = 50-2×3 = 44 and z = (44-50)/3 = -2. A lower-band touch is an observation, not a guaranteed reversal; a mean-reversion trade needs evidence, sizing, and costs.

13 / QF2-D3

Daily updates capture fresher information but may spend more on spreads and impact than the additional alpha they capture. Compare net returns at feasible liquidity and execution timing; a slow rule also risks holding stale positions.

14 / QF2-D4

Initially 0.06√100 = 0.60. After expansion 0.03√400 = 0.60. The simplified IR is unchanged before considering any change in costs, constraints, or reliability.

15 / QF2-D5

Gross = 160% + 60% = 220%; net = 160% - 60% = 100%. It violates the gross cap despite being net 100%.

16 / QF2-D6

1.5%-2.0% = -0.5 percentage points, all else equal. Gross outperformance does not establish investor value.

17 / QF2-D7

The individual trade has -2 bps expected net edge; the slower annual rule has 4% estimated net alpha. Compare matched horizons and cost conventions rather than ranking by gross forecasts alone.

18 / QF2-D8

No. The $0.07 adverse post-fill movement exceeds the $0.04 initial saving by $0.03 under the stated averages. This is evidence of adverse selection; compare completion, urgency, costs, and consistent markout benchmarks before choosing execution.

19 / QF2-D9

Covered-call profit = min(S_T,110)-100+4. At $105: $9; at $130: $14; at $0: -$96 per share. The premium cushions downside by $4 and caps upside at $14.

20 / QF2-D10

Check timestamps, freshness thresholds, and independent feed agreement; pause or escalate trading through a defined fallback when inputs are unreliable. Valid format and high model confidence do not establish current, trustworthy data.

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Source-key cautions and interpretation

The published key is evidence about the supplied pool, not a guarantee that every label and rationale agree. The Study Lab preserves published selections. The cases below distinguish those selections from the conceptual rule.

Known source conflicts

ML2-D10G2Q3 E | Reverse-key contradiction

ML2-D10G2Q3 / pool p. 571

Source statement: Adding channels changes the available input information but guarantees neither generalization nor causality.

Published selection: True

The stem asks for True when a claim is incorrect and False when it is correct. Statement E correctly says that adding channels changes available inputs but guarantees neither generalization nor causality. The official key nevertheless marks True and calls it incorrect, while its own rationale says more inputs can help or hurt and the architecture does not establish causality. The rationale supports the statement, so the logically consistent reverse selection is False. Preserve official True for pool scoring and distinguish it from the conceptual answer.

Conventions to keep explicit

Q-learning and episodes. A true terminal state has no bootstrapped continuation value. A time limit used to cut a continuing simulation may need a different treatment. Repeated episodes reuse experience; they do not manufacture independent market histories.

Exploration. Uniform exploration can include the greedy action. Compute action probabilities from the actual sampling rule, the number of legal actions, and any tie-breaking rule.

Breadth and transfer. The Fundamental Law is an approximate framework. Count effective independent decisions, align the evaluation horizon, and distinguish forecast skill from the ability to implement it under costs and constraints.

Costs and turnover. A cost quoted in basis points applies to a stated notional base. One-way turnover, total traded dollars, and complete round trips are different conventions. Avoid double-counting a spread already embedded in execution prices.

Options. Separate payoff from profit, strike from premium, and moneyness from profitability. State exercise style, dividends, financing, contract multiplier, and expiry assumptions before using parity or breakeven.

Source pages. A source question can continue onto later pages. The group and example citations identify its actual 1-based PDF start page, including the page corrections made during extraction review.

CS7646 / EXAM 2

Coverage and final readiness check

Scope and evidence

This companion covers ML Part 2 and quantitative finance Part 2 in the supplied ML4T Exam Question Pool, Rev081020261230: 420 question IDs, 140 groups, and 20 domains. Every group and every worked example in the reviewed Exam 2 teaching data appears in the domain chapters. Source scope is pp. 421-581 for ML2 and pp. 582-753 for QF2.

Question references

ML2-D3G2Q1 means machine learning, Part 2, domain 3, group 2, question 1. QF2 uses the same structure for quantitative finance. “Pool p.” refers to the original source PDF, not this guide. The chapter directory, PDF bookmarks, and footer provide navigation within this guide.

How this guide relates to practice

The chapters explain mechanisms and common distractors. Worked examples adapt or illustrate cited scenarios; they are teaching exercises rather than reproductions of every option in the source. The 20 self-test questions are derived retrieval practice, one per domain. Use the full Study Lab question bank to practice exact wording and the retained published labels.

Six calculation areas

The Study Lab’s calculator practice covers Q-learning updates, epsilon-greedy action probabilities, technical indicators, the Fundamental Law, trading costs, and option payoff/profit. Reproduce each calculation on paper and explain its assumptions before relying on a calculator output.

Ready when you can explain all four

1. Mechanism. Describe what changes after the action, why a signal might work, or which contract right creates the payoff.

2. Arithmetic. Write the formula with units, dates, sign, horizon, and denominator before substituting.

3. Qualification. State what the result does not prove: convergence, causality, independent breadth, executable fills, or positive net performance.

4. Wording. Translate the evaluated claim into the requested True/False convention and recognize the flagged source-key conflicts.

No official exam deadline is asserted here. The guide and planner are preparation tools, not course scheduling announcements.